Design and Manufacturing Project

Design, Construction, and Performance Analysis of a Pelton Hydraulic Turbine for Sustainable Electricity Generation

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Contents

1 Introduction

The purpose of this course was to give life to an idea we had. We were supposed to think about something and then create it without forgetting to make all the necessary studies. We decided to manufacture a Pelton water turbine to generate electricity. To better understand how a turbine works and to go a little further in fluid mechanics was our objective at the beginning. It was a good way to use our skills in SolidWorks and try to simulate the flow with Flow Simulation. At the end we wanted of course to give life to our idea and manufacture our project. We wanted one semester to try to finish the project, because one member of the group is an Erasmus student and won't stay after this semester. By the close of the year, the remaining members of the group finalized the project assignments. We tried to take our time for every step even if we were a bit under pressure and without forgetting that the most important thing was to learn something new and avoid the things that we already knew.

2 Processus

2.1 Study

Before directly starting the design and manufacturing we had to study and go deeper into the subject. First, we analyzed and tried to understand the functioning and the special shape of a Pelton turbine. It was useful to understand and analyses how it works, in what context is it used, what type of materials do they use to manufacture it, etc.… The first question we asked ourselves was: How does a Pelton Turbine work?

The concept is based on the conversion of mechanical energy into electricity. The water flow will put the wheel and the shaft in rotation which will start the rotation of the generator and generate electricity. This type of hydraulic turbine is still in use today, even though it was created in 1879. Many hydroelectric power stations use this system. Pelton wheels are usually built in a single piece and are cast using ferrous materials with high chromium and nickel alloys. A huge problem in that kind of installation is the cavitations which can break the whole turbine.

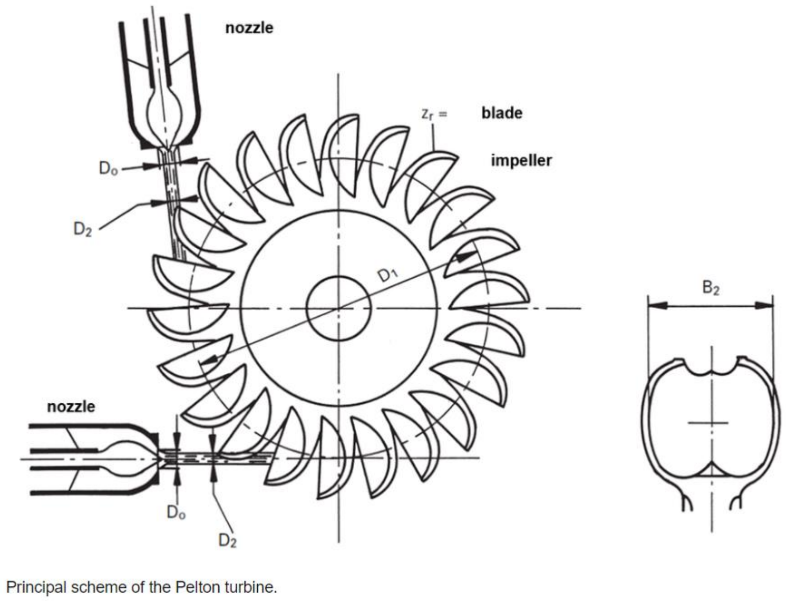


Figure 1: Representation of a Pelton Turbine

The picture above is a representation of the whole system Nozzle/Turbine, in this case two nozzles.

The whole system is composed of the following different parts:

All the drawings are available as PDF and attached.

2.1.1 Pelton Wheel

It has its recognizable “spoon” shape which makes it so unique. It has been invented and improved over the years. The reason for it is the following: This shape is the most appropriate one to use the kinetic energy of the water. The jet splits in two and transfers a large part of its kinetic energy. The water then flows sideways and can be reused or discharged. It is interesting to know that this shape has been discovered thanks to a lot of different tries. It allows a rate of about 90 %. The number of troughs is based on a formula derived from manufacturers' experience, i.e., Z=15+D/2d, where Z = number of troughs, D = original diameter, d = diameter of the water jet. In practice, their number varies from 15 to 24. The diameter of the center of the buckets is called the Ø Pelton, a diameter used to position the axis of the water jet, which must be tangent to this Ø Pelton. In our case we have less because we don’t have huge waterflow like in the industry.

2.1.2 The shaft

The shaft of a Pelton turbine is a critical component that transfers rotational energy from the turbine runner to an external device, such as a generator. It's typically made of strong materials like steel to withstand high rotational speeds and torque. The shaft needs to be precisely engineered to ensure smooth operation and minimize vibration, which could lead to mechanical failure. Regular maintenance and inspection of the shaft are essential to ensure the efficient and safe functioning of the Pelton turbine system.

2.1.3 The generator

It is the only piece of our system that will not be built by us. It is the piece which will convert our kinetic energy into electricity. It will be connected to our shaft thanks to a transmission system.

2.1.4 The support

The turbine needs support and we decided to build our own. It will be the basis of the system and directly connected to almost everything. It is useful to control the evacuation of the water, and in real life protect the turbine from disturbance variables.

2.1.5 The nozzle

The nozzle of a Pelton turbine is a crucial component that directs high-pressure water jets onto the buckets of the turbine wheel. It's designed to control the flow and angle of the water jets, optimizing the turbine's efficiency. The shape and size of the nozzle determine the velocity and impact of the water on the turbine blades, influencing the power output of the turbine. Adjusting the nozzle allows for efficient operation across varying water flow rates and head pressures.

With the nozzle we will be able to aim directly and exactly where we want on the throughs. It is an important piece which will enable us to put the turbine in rotation. In fluid mechanics we have learned that we can calculate the speed of water with $v=\sqrt{2gh}$. It is working in the case where we are dealing with a waterfall at a low speed, so the equations of Bernoulli simplified themselves. In our case we cannot consider the speed equal to 0 at the beginning, but if we ignore the friction, we could use the Konti Equation. $A_1c_1 = A_2c_2$ (it’s just water so the density doesn’t need to appear in the equality). We utilized a standard household water tap and conducted online research to gain insight into velocity.

2.1.6 Ball bearings

That was one of the biggest challenges of our project. Manufacturing working ball-bearing seems easier than it is. The purpose of them is to make the rotation of the shaft through the support possible.

2.1.7 Transmission system

The transmission system is the part of our project which makes the transmission between the shaft and the generator possible. Every part of our system has its own problematics, the way to connect every part of our system was a challenge as well. At the beginning we thought about two options, Gears or a belt. Gears seems interesting because we could improve the rotations per minute and generate more electricity. We have manufactured the gears but at the end we did not use them because the ball bearings were way too loose, and we were not able to transmit the energy. With more time we would have thought on a better solution, or we could have manufacture others ball bearings. And we use adapter system for transmission.

2.1.8 Biomimicry

In our project, biomimicry played a crucial role in the design of the Pelton turbine's wheel and bucket configuration. Inspired by natural structures found in organisms such as shells or seed pods, nautilus shell, our team implemented a design featuring distinct separator elements between the bucket segments. This biomimetic approach not only enhances the efficiency of water energy capture but also ensures optimal flow dynamics, closely resembling the functional separation observed in certain natural structures. By integrating these biomimetic principles into our turbine design, we have achieved a more sustainable and efficient solution for electricity generation. "This emphasizes how your turbine's design directly reflects and is inspired by natural structures, particularly in the unique separator elements between the bucket segments.

2.2 Design

Then after learned everything we needed and having done all the necessary research, we decided to start the designing of our project. We used SolidWorks because everyone was more or less used to it. It was also a way to learn more about it and use our skills in practice. We started with the following design:

Sketched on SolidWorks, this was our first "prototype". It is based on one shaft, the wheel, two ball bearings and the support. It is relevant to say that we kept almost nothing out of this prototype for our final version.

This first prototype was without any generator and transmission system, it was a way to see how it would look like and understand what could works and what not. We never built this one in the real life.

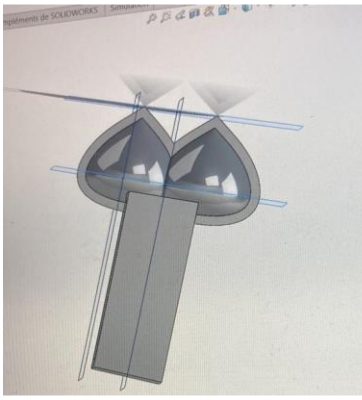
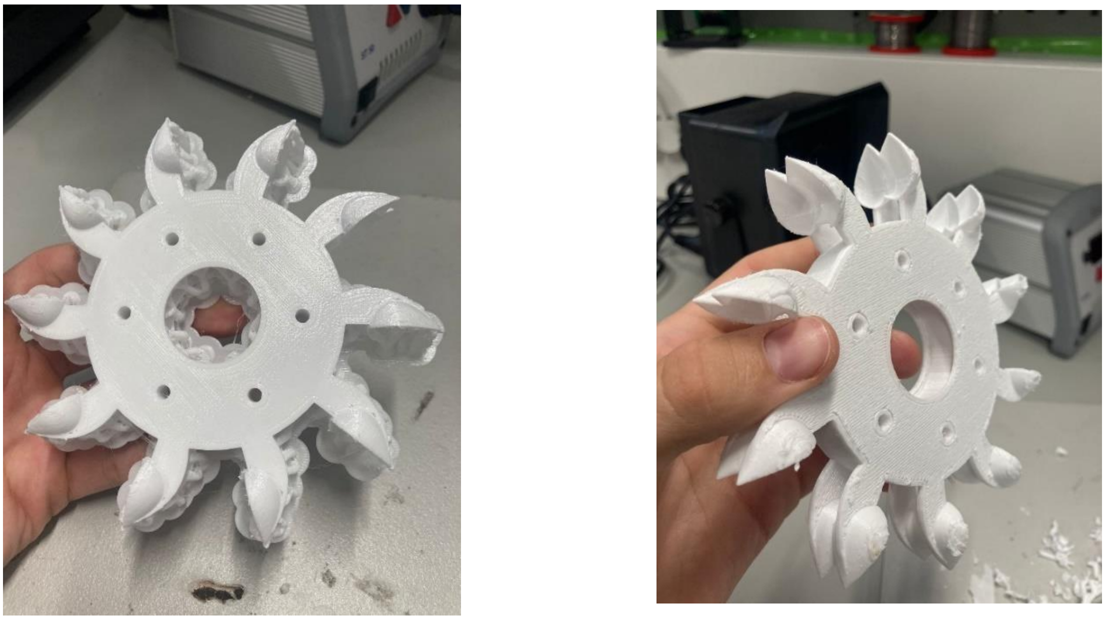


Figure 2 troughs designed with shell.

Figure 3 Second Wheel Prototype

2.3 Manufacturing

After many tries, we went through the “final” version of our turbine. We decided to manufacture this prototype to see what could have been wrong and what would work. Sometimes trying through the experiences is the best way to understand. The easiest solution and the cheapest was to use a 3D printer. We went to the JHC and even if one of us already knew how it works, it was a great possibility for the others to learn more about it. We began with the wheel, which is the most important piece, after the first printing, we were quite delighted of the result. The only one problem is that its throughs were too thin. So, we changed it and continued with other pieces.



Manufactured Wheel before/after cleaning of the “organic” supports.

2.4 Testing and calculations

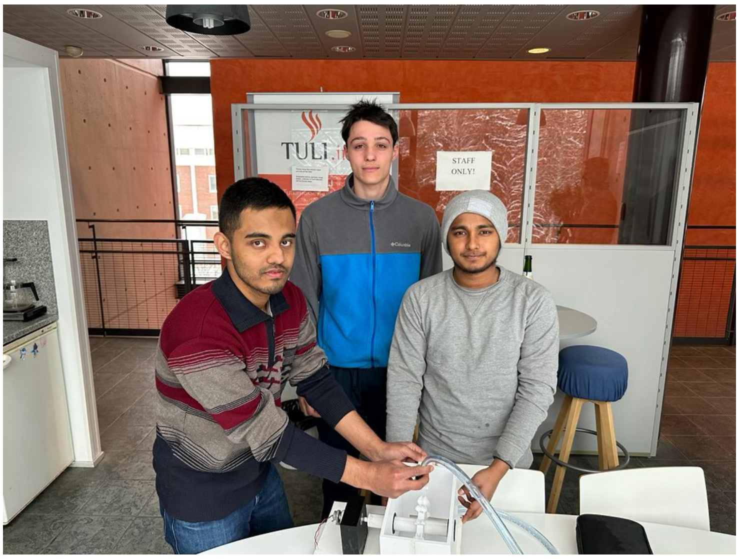
After our first prototype, it was time for testing. You can watch our first testing through that video.

https://youtube.com/shorts/CjzsIy845N0?si=LFTvkuE9R1A1NGQM

We were a bit naive about the transmission system and we really thought that the hardest part of our project was finished. We used adapter transmission system.

Before, an insight of the transmission system we tried to implement with gear:
https://youtube.com/shorts/BoIkWcFZWAU?feature=share

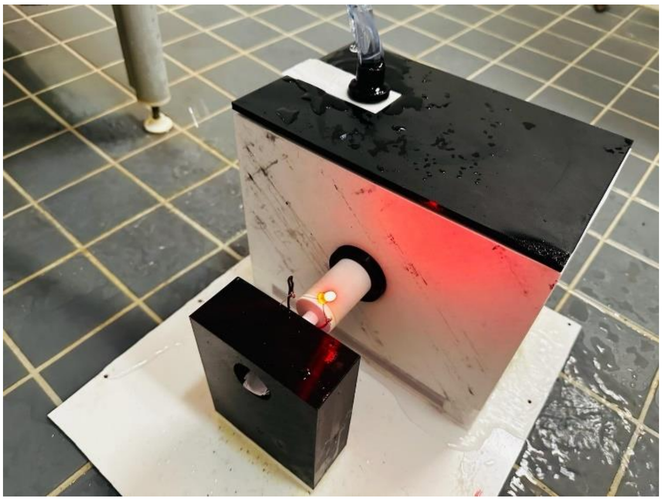
In our initial testing phase during the first half of the academic year, we endeavored to produce electricity. This decision was influenced by the participation of an exchange student who would not be able to remain for the entirety of the academic term. During that period, our attempts to generate electricity were unsuccessful, and we encountered several limitations which we observed. Then by time we learned and did product development and at the end of the year we finally succeed to generate electricity. Over time, we gained knowledge and engaged in product development, eventually achieving success in generating electricity by the end of the year.



Picture of the group (without Rajiv who was not here) with the Turbine which was our first testing.

This is the link to our conclusive electricity generation test:

https://www.youtube.com/watch?app=desktop&v=r5DEqzfE4aQ



Picture of the turbine while its generating electricity

Picture of the group (Rajiv and Farhan is here) with the Turbine which was our final testing.

We tried to calculate the rotation per/minute thanks to what Lucas had learn during his Fluid mechanic lecture.

RPM and Angular speed:

We have RPM = $\frac{v*60}{D*\pi}$

with, D = Diameter of the Pelton wheel

V = velocity of the jet water

RPM = rotations per minute

We got D = 80 mm

v = 0.5 m/s for a normal tap

RPM = $\frac{0.5*60}{0.08*\pi}$ = 119,3 RPM

Our calculations are probably a good approximation because if we put the video in slow motion and count every rotation, we have around the same number of rotations.

Then we can calculate the speed of the turbine:

$\omega=\frac{30*n}{\pi}$ with n = RPM in rotations per minute

w = angular speed in rad/s

$$\omega = \frac{30*119{,}3/60}{\pi} = 18 \text{ rad/s}$$

This speed looks completely possible and could be implemented in a real installation.

Source: https://en.wikipedia.org/wiki/Rotational_frequency

Whirling of shaft. (critical speed)

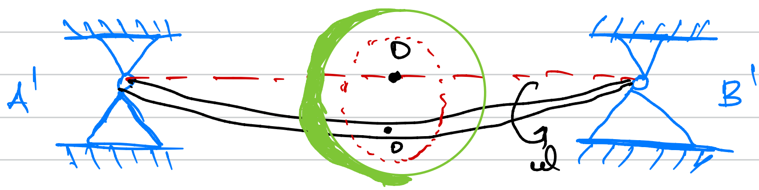
① When the natural frequency of the system coincide with the external forcing frequenc it is called resonance. The speed at which resonance occurs are known as the critical speed or whirling speed.

② At these speeds the amplitude of vibration of rotor is excessevely larg and the large amount of force is transmitted to the foundation or bearing.

③ In the region of critical speed the system may fail. therefore it is very important to find the natural frequency of the shaft to avoid the occurance of critical speed.

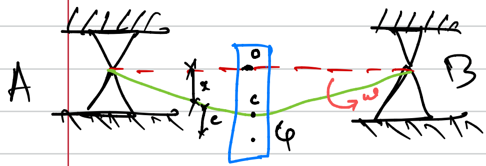
④ Turbines compressors, electric motors and pumps usually incorporate a heavy rotor mounted on a light flexible shaft that is supported in bearing.

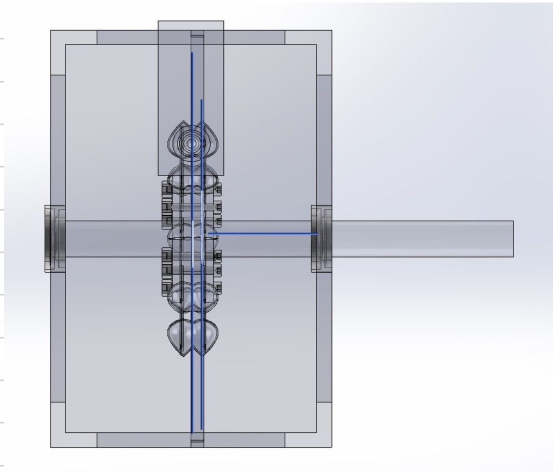
⑤ Centre of mass of there mounting does not coincide with the centre line of shaft. Thus shaft rotates a centrifugal force act on it which makes the shaft be bend in that direction of ecentricity of the centre of mass.



Now,

Let us consider a shaft rotating horizontally in bearing A and B as shown in figure





Let,

as assume shaft being of negligible weight carries a rotating disc of weight (W).

k_t = stiffness of shaft

e = eccentricity of the disc = C·G

x = lateral deflection of shaft from (O)

ω = uniform angular velocity of shaft.

ω_c = critical speed of shaft.

C = Geometric centre of the disc.

G = centre of gravity of the disc which is at a distance (e) from (C).

Now,

Consider the equilibrium of the disc two force are acting on it.

1) The centrifugal force - outward (radially) → G
2) The restoring force - radially inward → C

∴ For equilibrium these forces acts on same line O, C, G

$k_t * x$

As we know,

Restoring Force = centrifugal Force. $\rightarrow$ $(m\omega^2(x+e))$ $r = (x+e)$

$\left(m = \frac{w}{g}\right)$

$$k_t x = \frac{w}{g}\omega^2(x+e)$$

$$k_t x - \frac{w}{g}\omega^2 x = \frac{w}{g}\omega^2 e$$

$$\left(k_t - \frac{w}{g}\omega^2\right)x = \frac{w}{g}\omega^2 e$$

$$\therefore \frac{x}{e} = \frac{w/g * \omega^2}{k_t - \frac{w}{g} * \omega^2}$$

$$\frac{x}{e} = \frac{1}{\frac{k_t * g}{w * \omega^2} - 1}$$

Also we know that, $\left[\omega_c^2 = \frac{k_t g}{w} = \frac{g}{\delta}\right.$ $(\because \delta$ is static deflection$)$

So,

$$\frac{x}{e} = \frac{1}{\left(\frac{\omega_c}{\omega}\right)^2 - 1} \quad \text{eq. 1}$$

When,

$\omega_c = \omega$ [$\frac{x}{e}$ ratio is infinity so the particular solution of critical speed is

$$\boxed{\omega_c = \sqrt{g/\delta}}$$

Now,

$\left[\omega_c = \sqrt{g/\delta}\right]$

we know that

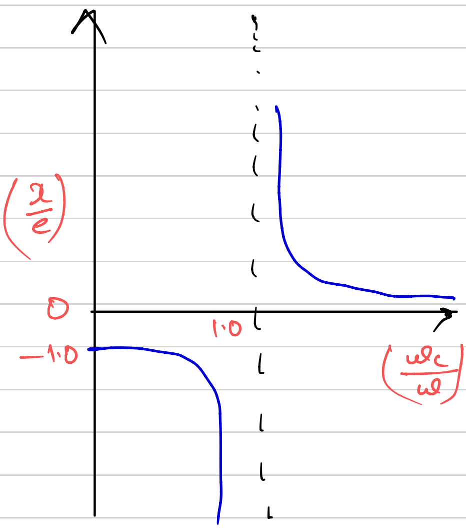
$\omega_c = 2\pi f_c$

$$\boxed{\therefore f_c = \frac{1}{2\pi} * \sqrt{g/\delta}} \text{ Hz}$$

With the help of equation (1)

$\omega = \omega_c$.

the amplitude ratio become infinity which causes severe vibration and excessive load on bearing and the disc will try to fly out.



So proper damping is provided to control the disc from flying out.

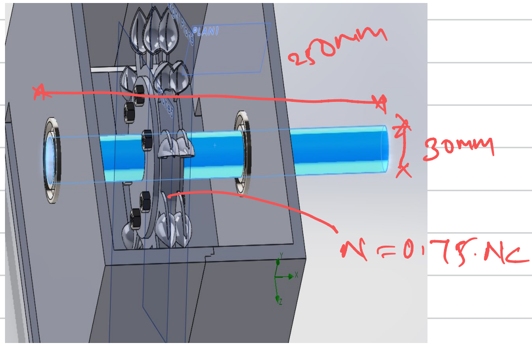
$\therefore \omega < \omega_c$, the amplitude ratio will be positive and C will lie between 0 & G

$\therefore \omega > \omega_c$, the ratio of (x/e) is found to be (-ve) it means (G) falls between 0 and c, thus at high speed the disc rotates with heavy side inside means at (C & G)

$\therefore$ Then the disc rotates about its C & G. (G) and vibration are minimum.

(calculation of our shaft for project.)

As per our project we have a shaft of 30mm diameter and 250mm long is supported with long bearings at it's ends.



Given,

lets assume the given details.

Diameter(d) = 30mm

Length (L) = 250mm

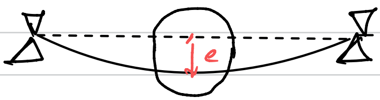
(N) = 0.75 Nc

e = 0.25 mm

E = 200 * 10^9 N/m^2

[maximum bending stress when the the shaft is rotating at 75% of the critical speed]

Now,



we know,

$$\delta = \frac{wL^3}{192EI}$$

where,

$$I = \frac{\pi}{64} d^4$$

$$= \frac{\pi}{64} * (0.03)^4 = 39.7 * 10^{-7} m$$

$$\therefore \delta = \frac{50 * 9.81 * (0.25)^3}{192 * 200 * 10^9 * 39.7 * 10^{-7}}$$

$$= 5.02 * 10^{-8} m.$$

So we know that,

$$\text{Natural frequency } (f_c) = \frac{1}{2\pi}\sqrt{g/\delta}$$

$$= \frac{1}{2\pi} * \sqrt{\frac{9.81}{5.02 * 10^{-8}}}$$

$$= 2224.84 \text{ Hz}$$

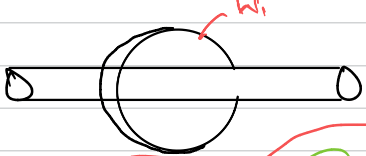
$$\boxed{2224.84 * 60 = 133491.62 \text{ RPM}}$$

Now,
calculating the Bending stress
we know,

$$M = \frac{W_1 * L}{8}$$

[W_1 = carrying a point load at the centre. (additional weight)]

$$\therefore \frac{\sigma I}{d/2} = \frac{W_1 * L}{8}$$



$$\therefore \frac{M}{I} = \frac{\sigma}{y}$$

$$M = \frac{\sigma I}{d/2}$$

So

$$W_1 = \frac{\sigma I * 8}{L * d/2}$$

$$= \frac{\sigma * 39.7 * 10^{-7} * 8}{0.25 * \frac{0.03}{2}}$$

$$= 8.46 * 10^{-3} \sigma$$

Now,

If Additional deflection due to the load (W_1)

$$\therefore y = \frac{W_1}{W} \delta$$

$$\left[\frac{\delta}{W} = \frac{y}{W_1} \right]$$

$$= \frac{8.46 * 10^{-3} \sigma}{50 * 9.81}$$

$$= 1.72 * 10^{-5} \sigma$$

$\therefore$ Natural Frequency

$$f_c = 133491.62 \text{ RPM}$$

NO22EL

Bernoulli equation

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$$

$\downarrow$ Pressure $\quad$ $\downarrow$ Kinetic $\quad$ $\downarrow$ Potential

P = Pressure

ρ = density

V = Velocity

g = gravity

h = ~~high~~ height

$Q = AV$

continuity equation

$$A_1 V_1 = A_2 V_2$$

A = ~~Velocity~~ Area

V = Velocity

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$$

$P_1 = 63486$ Pa

$v_1 = 9.9$ m/s

$v_2 = 0.7$ m/s

$\rho = 1000$ kg/m³

$H_1 = H_2$

$$P_1 + \frac{1}{2} v_1^2 \rho_1 + \cancel{\rho g h_1} = P_2 + \frac{1}{2}\rho v_2^2 + \cancel{\rho g h_2}$$

$$\Rightarrow 63486 + \frac{1}{2} \times 1000 \times (9.9)^2 = P_2 + \frac{1}{2} \times 1000 \times (0.7)^2$$

$$\Rightarrow 112491 = P_2 + 245$$

$$\Rightarrow P_2 = 112246 \text{ Pa}$$

Head,

$$H = \frac{P}{\gamma}$$

$$= \frac{112246}{9810}$$

$$= 11.44 m$$

Ⓐ $P = 112246$ Pa

Input Power,

$$P_i = \gamma Q H$$

$$= 9810 \times 8.8 \times 10^{-4} \times 11.44$$

$$= 99.32 \text{ watt}$$

Output Power,

$R = 80$ mm
$= 0.08$m

$$P_o = T \times g \times \left(\frac{2\pi N}{60}\right) \times R$$

$$= 1.36 \times 9.81 \times \frac{2\pi\, 119.3}{60} \times 0.08$$

$$= 13.32 \text{ watt}$$

Efficiency $\eta = \frac{P_o}{P_i}$

$$= \frac{13.32}{99.32}$$

$$= 13\%$$

Scanned with CamScanner

2.5 Simulation Result

Flow Simulation With SolidWorks

Water comes through the water pipe and flows through the nozzle. Now here is the flow simulation in the nozzle.

Flow Simulation in the Nozzle

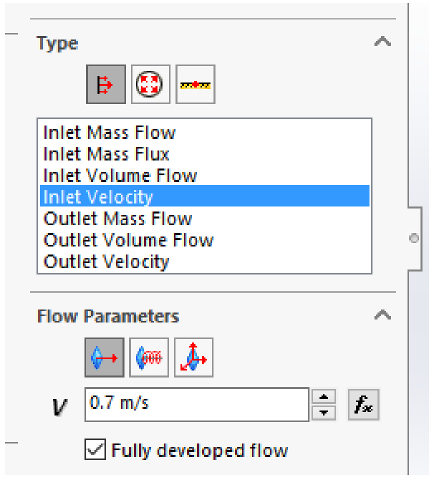
Before the simulation, This data putted in the system to see the simulation:

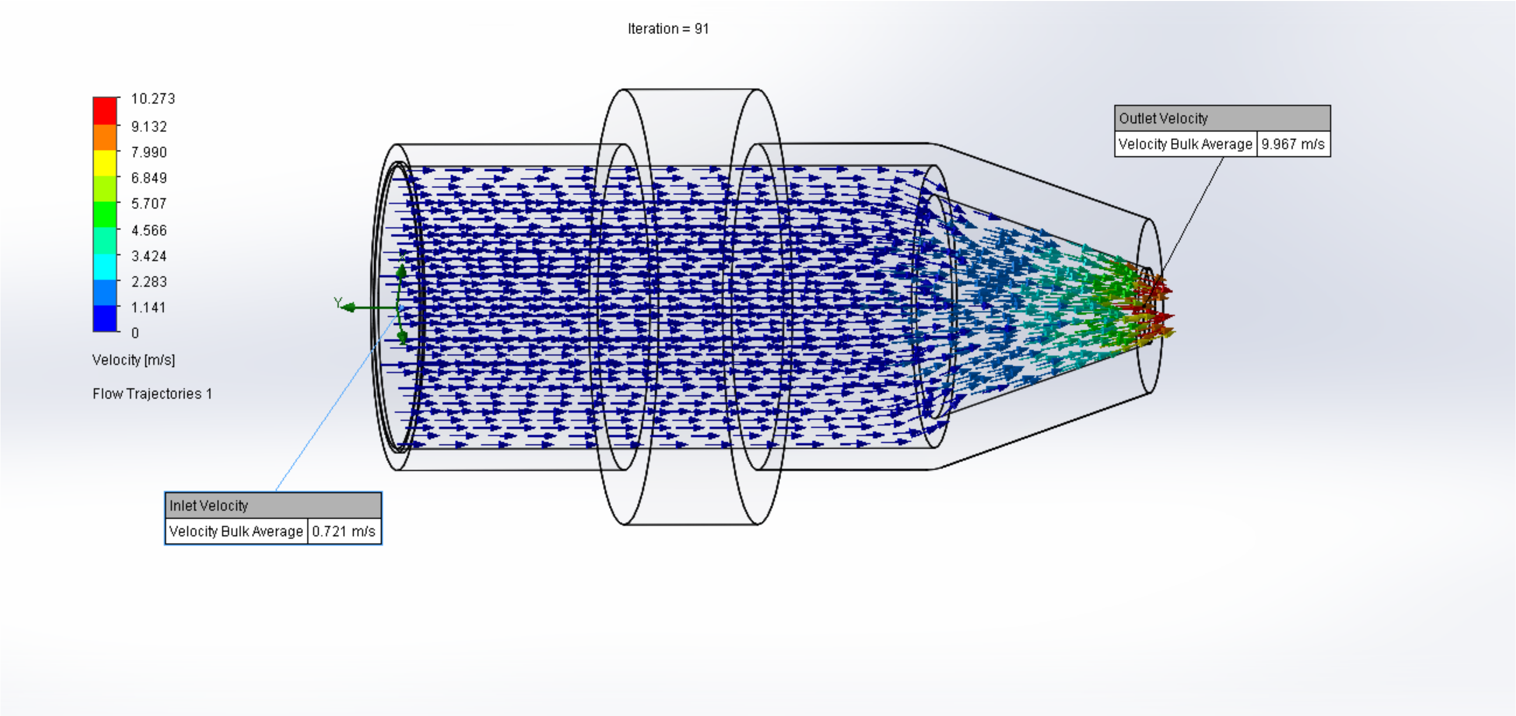
- Gravity: 9.8 m/s^2 which is working on –y direction.
- Temperature: 293.2 k
- Pressure: 101325 Pa
- Mesh level 06
- Flow type: Laminar
- Fluids: Water

Normal water tap velocity of Tap water (low noise) is 0.5 to 0.7 m/s and Tap water is 1.0 - 2.5 m/s.

Source:
http://ndl.ethernet.edu.et/bitstream/123456789/89817/1/Maximum%20Flow%20Velocities%20in%20Water%20Systems.pdf

So, Lets test both with low and high flow.





First, Simulation with 0.7 m/s inlet velocity.

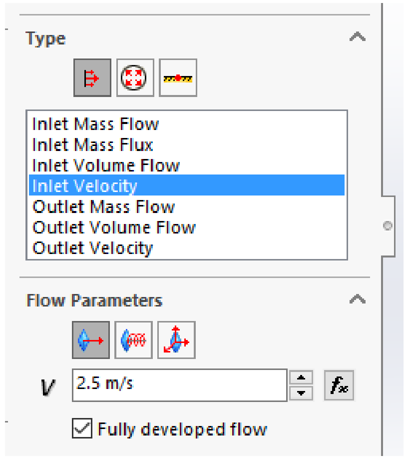
Simulation Result: Water enters the nozzle with a velocity of 0.7 m/s and exits with a velocity of 9.96 m/s. Maximum velocity can be 10.3 m/s and minimum velocity can be 5.6 m/s.

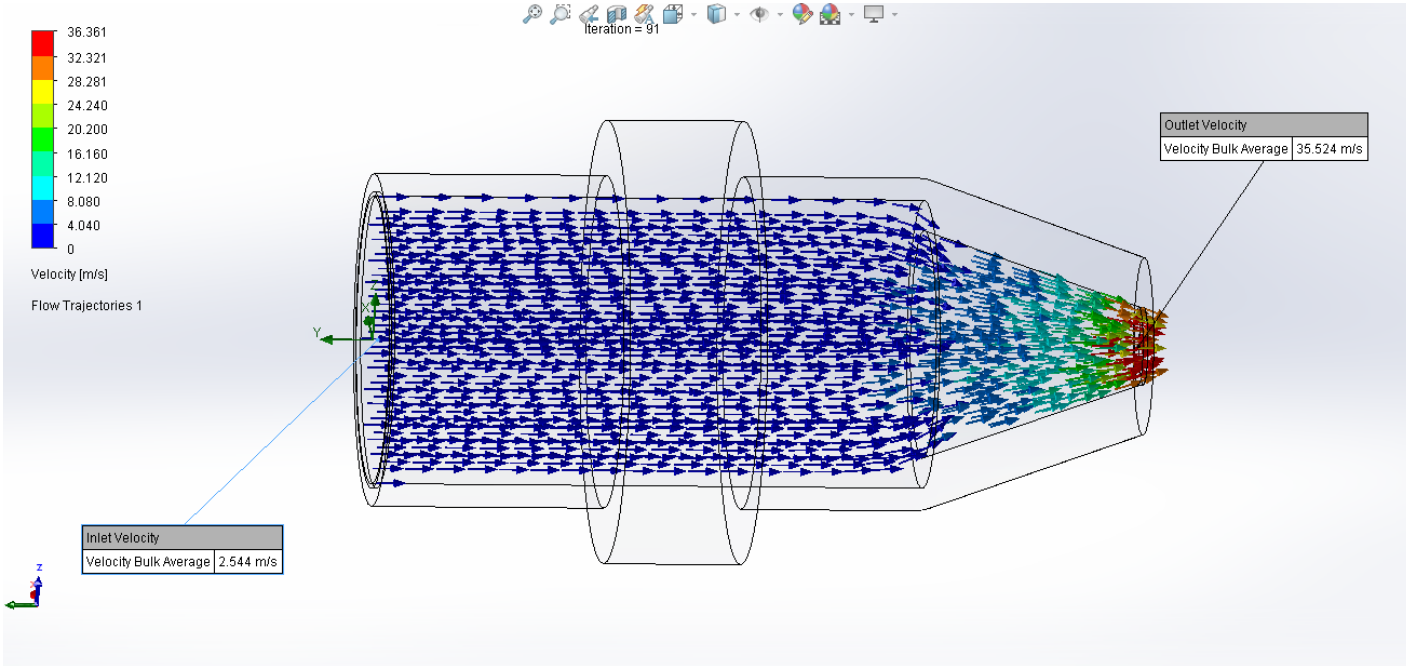
Our target was to increase the velocity of the water coming out of the water source and we succeeded in that.

Note: Here water flow and gravity force is working on same direction.

Outlet Velocity Result:

Local Parameter	Minimum	Maximum	Average	Bulk Average	Surface Area [m^2]
Velocity [m/s]	5.629	10.273	9.930	9.967	9.6476e-06





Second, Simulation with 2.5 m/s inlet velocity.

Simulation Result: Water enters the nozzle with a velocity of 2.5 m/s and exits with a velocity of 35.4 m/s. Maximum velocity can be 36.3 m/s and minimum velocity can be 23.8 m/s.

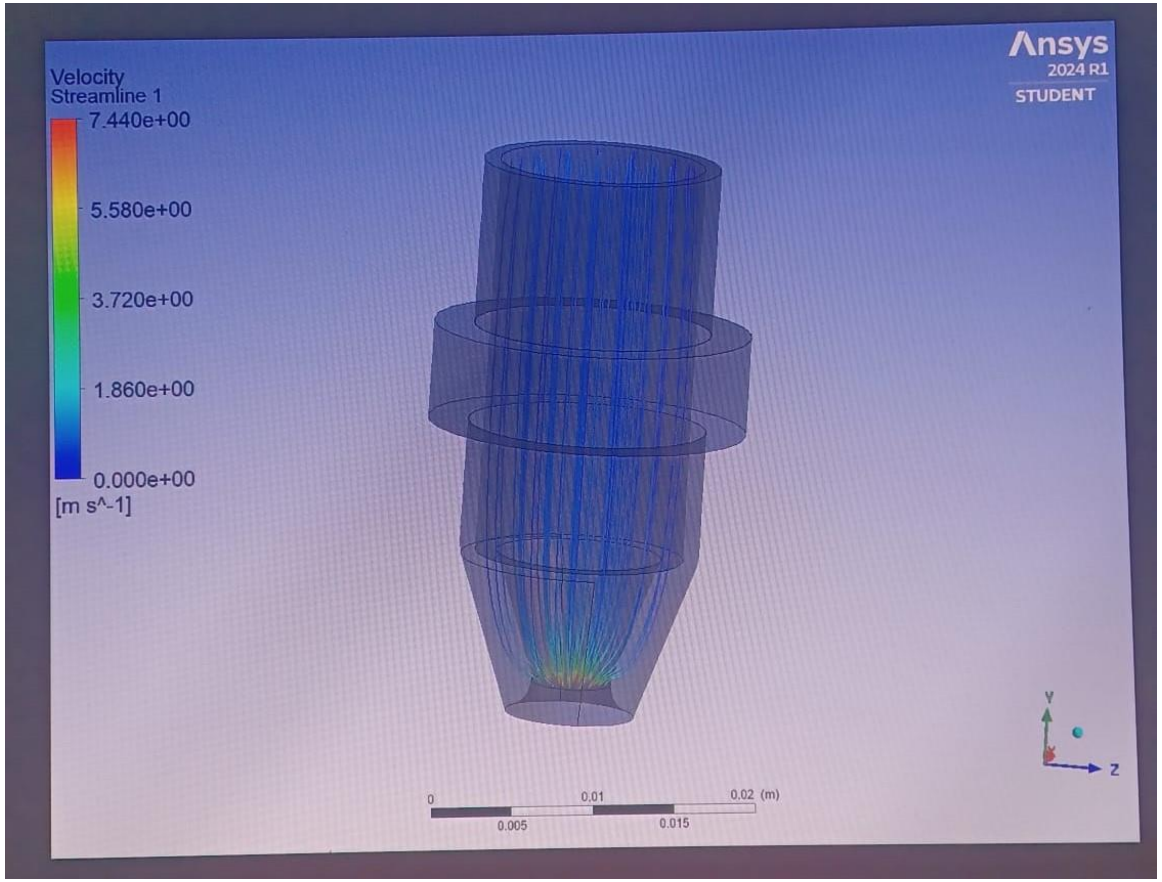
Our target was to increase the velocity of the water coming out of the water source and we succeeded in that.

Note: Here water flow and gravity force is working on same direction.

Outlet Velocity Result:

Local Parameter	Minimum	Maximum	Average	Bulk Average	Surface Area [m^2]
Velocity [m/s]	23.811	36.361	35.447	35.524	9.6476e-06

Flow Simulation With Ansys



We also test the nozzle it in the Ansys. Here we put inlet velocity 0.8 m/s. So, water enters the nozzle with a velocity of 0.8 m/s and exits with a velocity of 7.5 m/s.

In SolidWorks we got 9.9 m/s exit speed from inlet velocity 0.7 m/s. And in Ansys we got 7.5 m/s exit speed from inlet velocity 0.8 m/s.

The outcomes derived from two distinct software programs are in close in proximity.

Let's, Calculate how many **mass flow rate** is acting in the exit point of the nozzle:

Here,

- Diameter of the Nozzle exit circle, D = 3.4 mm
- Cross section area, A = π x 1.7^2 = 9.1 mm^2 = 0.0000091 m^2
- Density of water, ρ = 1000 kg/m^3
- Flow velocity of the water, v = 10 m/s

In the first case we got 10 m/s velocity, so we use this velocity here.

Then we have a formula,

Mass flow rate = ρ x v x A = 1000 kg/m^3 x 10 m/s x 0.0000091 m^2 = 0.097 Kg/s = 91 g/s

This implies that a flow of 91 grams of water per second is exiting the nozzle.

Source: https://en.wikipedia.org/wiki/Mass_flow_rate

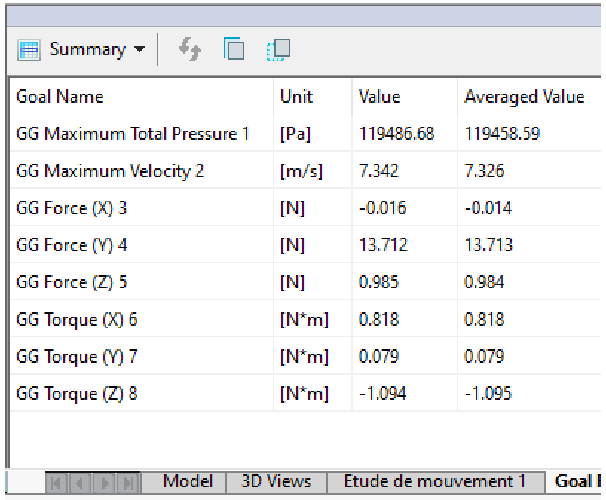
Water flows through the nozzle and falls into the wheel bucket. Now here is the flow simulation of inside the turbine.

Flow Simulation Inside Turbine:

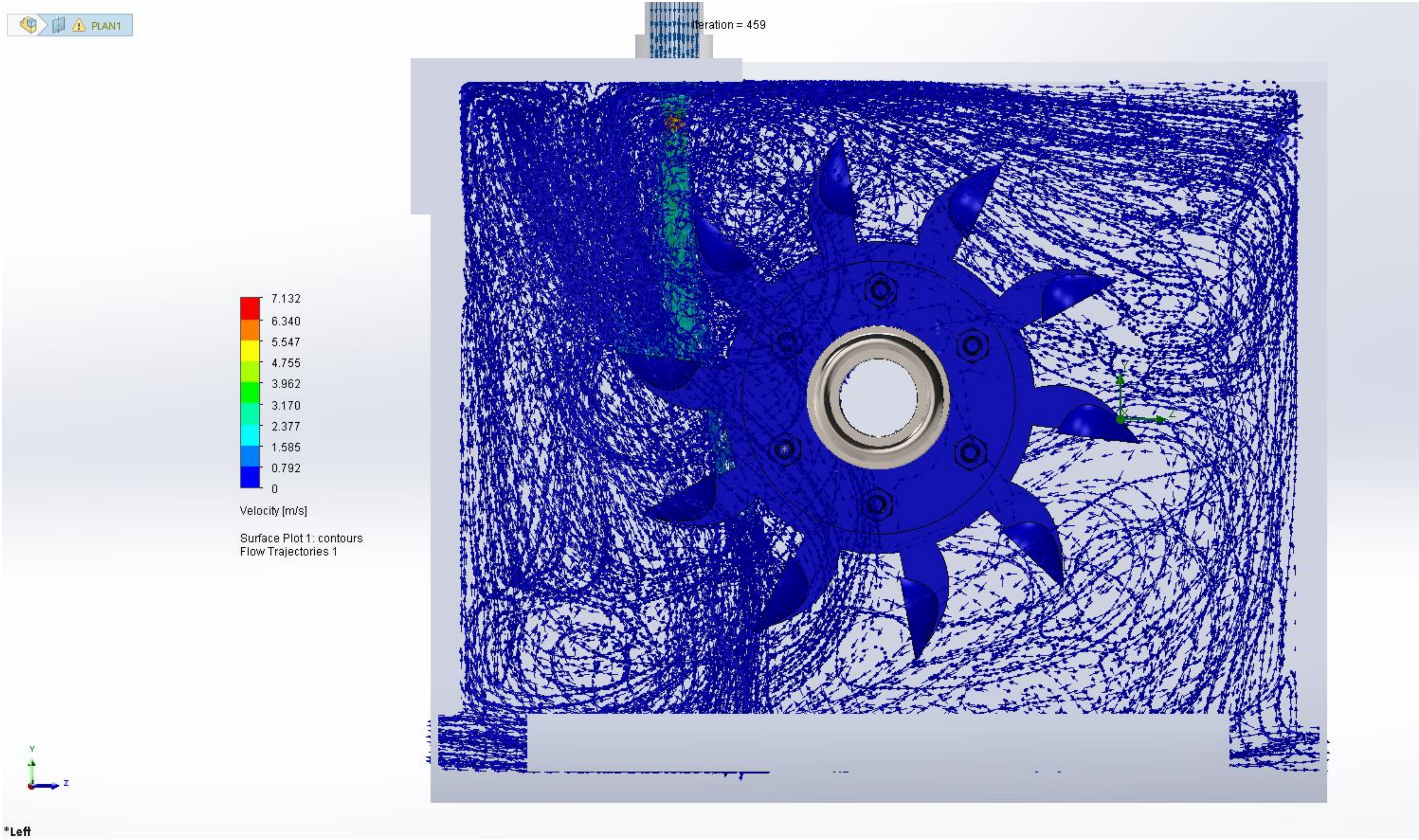
Before the simulation we put this data in the system to see the simulation:

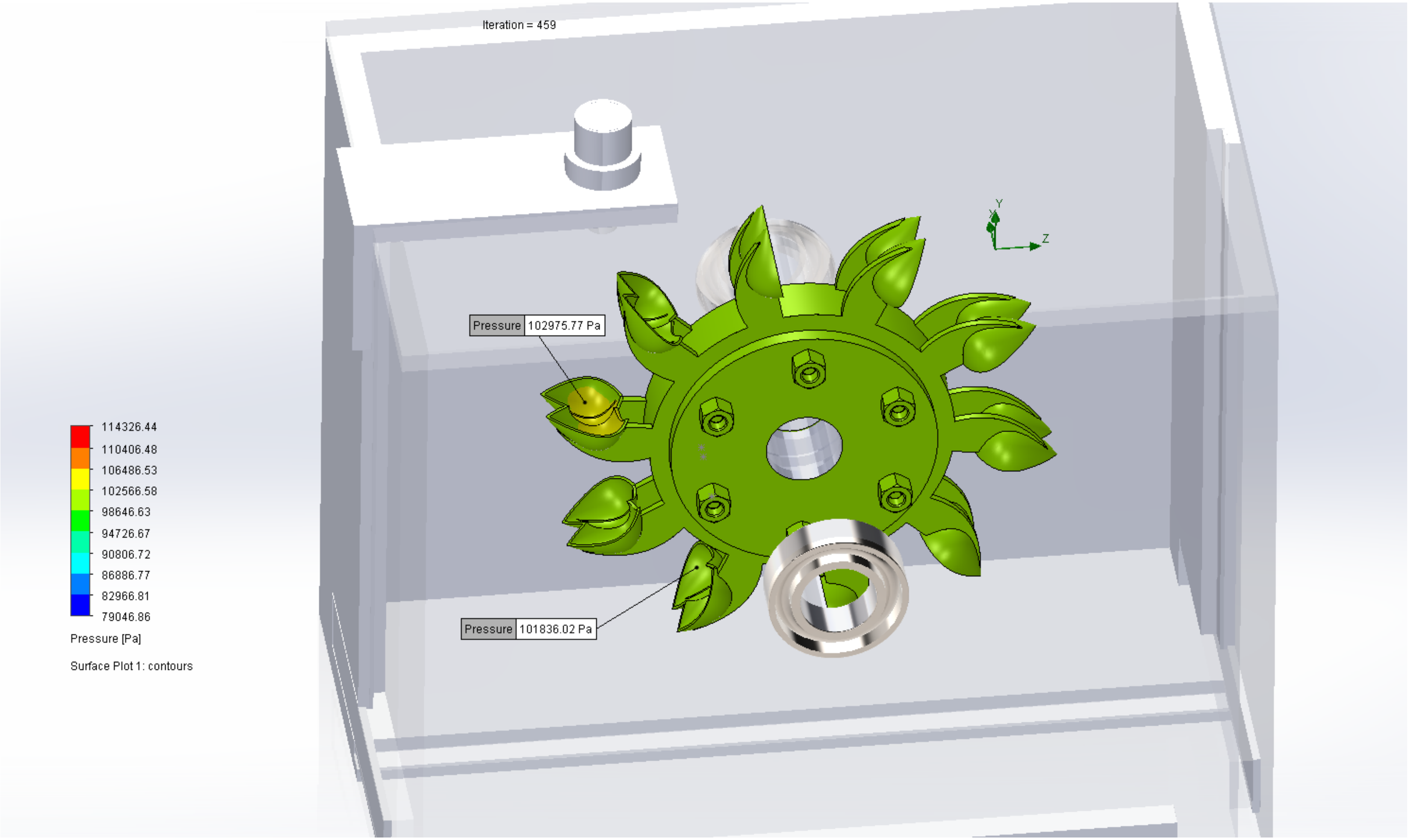
- Gravity: 9.8 m/s^2 which is working on –y direction.
- Temperature: 293.2 k
- Pressure: 101325 Pa
- Mesh level 06
- Flow type: Laminar
- Fluids: Water
- Water inlet velocity is 01 m/s

Simulation Result:



Goal Name	Unit	Value	Averaged Value
GG Maximum Total Pressure 1	[Pa]	119486.68	119458.59
GG Maximum Velocity 2	[m/s]	7.342	7.326
GG Force (X) 3	[N]	-0.016	-0.014
GG Force (Y) 4	[N]	13.712	13.713
GG Force (Z) 5	[N]	0.985	0.984
GG Torque (X) 6	[N*m]	0.818	0.818
GG Torque (Y) 7	[N*m]	0.079	0.079
GG Torque (Z) 8	[N*m]	-1.094	-1.095





Simulation Result: The pressure on the water falling bucket is 102975.77 pa.

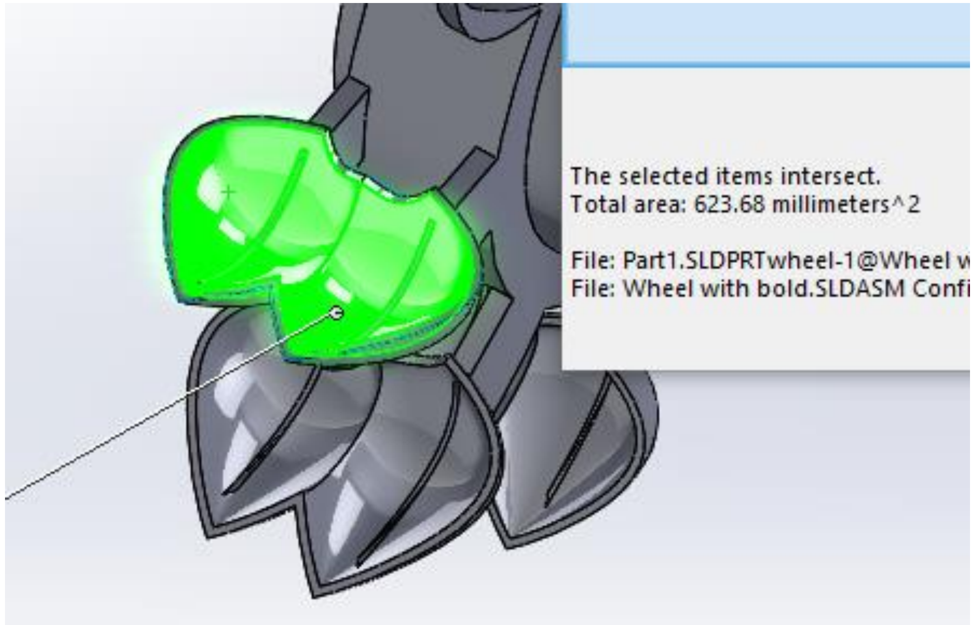
A succinct outline of the simulation.

Water is discharged from a nozzle, cascading into a wheel bucket. Within the bucket, it experiences the highest pressure, measured at 102975.77 Pa. Meanwhile, a second bucket positioned beneath registers a lower pressure at 101836.02 Pa. Consequently, a greater force exerts itself within the falling bucket. Following Newton’s first law, this discrepancy in force initiates a counterclockwise rotation of the wheel. We observe Newton's third law states that after the water flows into the bucket, it pushes the bucket downward and water rises up. Around the exit, we can also see some turbulence or water circulation.

Let's consider a scenario where the ball bearing fails and the shaft becomes stuck, resulting in the wheel bucket remaining motionless while water continues to fall onto it. Let's explore how the bucket would respond in such circumstances.

From the simulation we have seen the pressure on the falling bucket is 102975.77 pa. and we got the falling bucket area from the simulation is 623.68 mm^2.

Important: Despite the pressure not being distributed across the entire bucket, our analysis here pertains to the entire bucket.

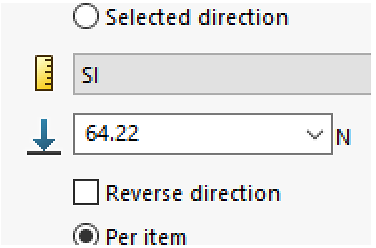


Now, we have pressure and area, so we can see how many force is acting here.

Force = pressure x area

$= 102975.77$ pa x $0.00062368\ m^2$

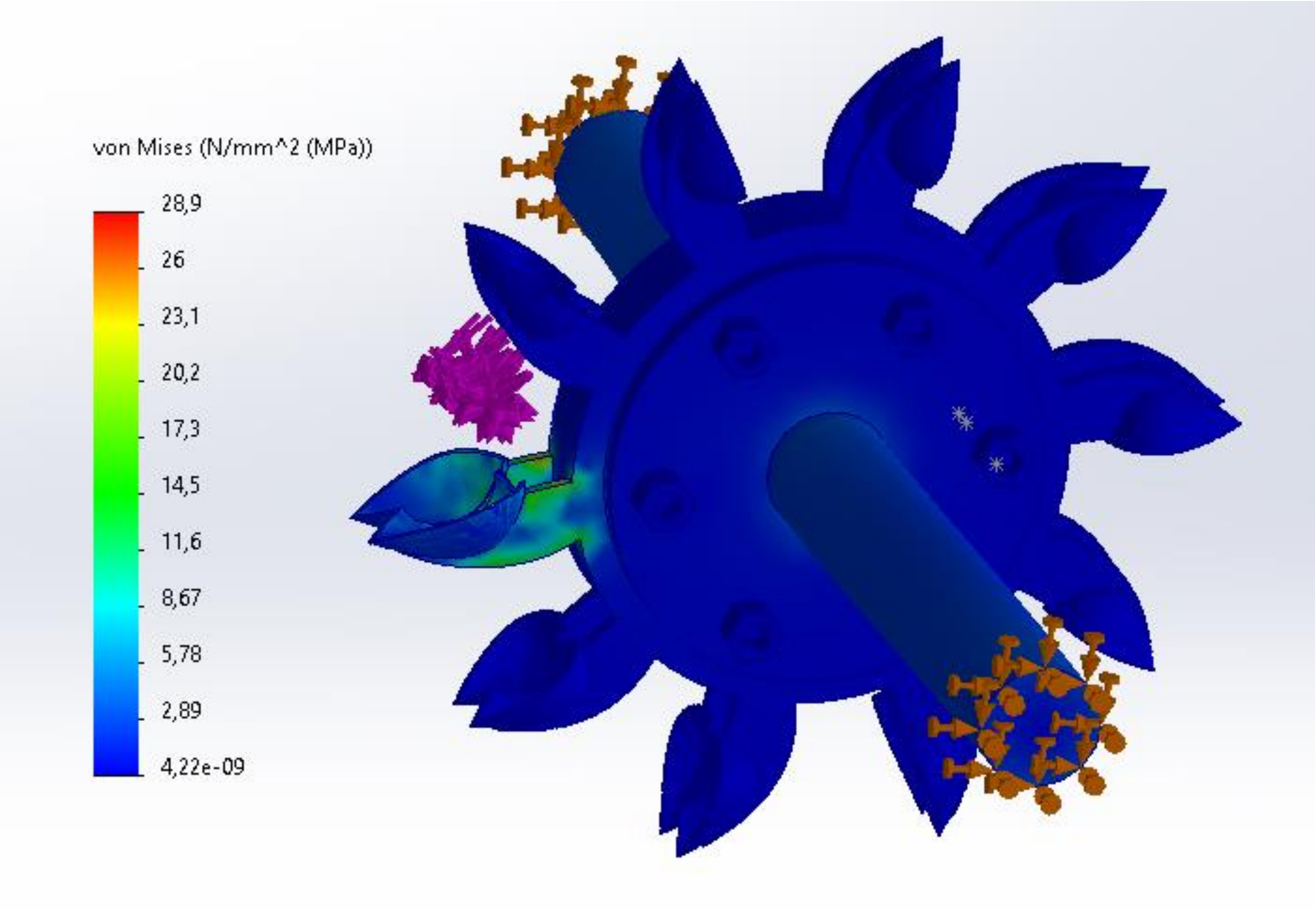
$= 64.22$ N



Let's examine the behavior of the bucket under these conditions.

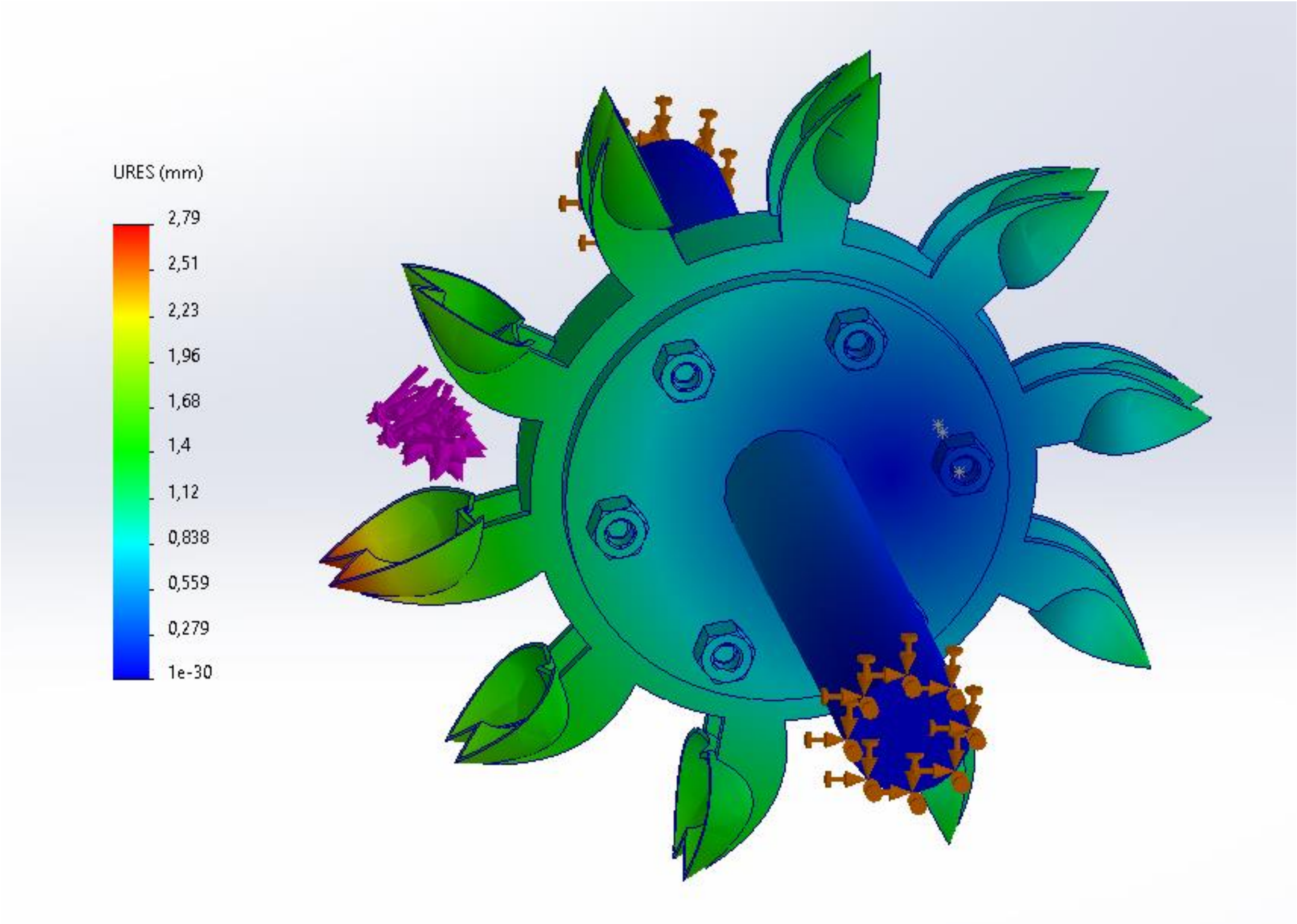
Before the simulation, This data putted in the system to see the simulation:

Material: ABS
Applied Force: 64.22 N



Static Simulation Result:

The maximum stress observed is 26 N/mm², concentrated at the bucket support (highlighted in orange) of the component.

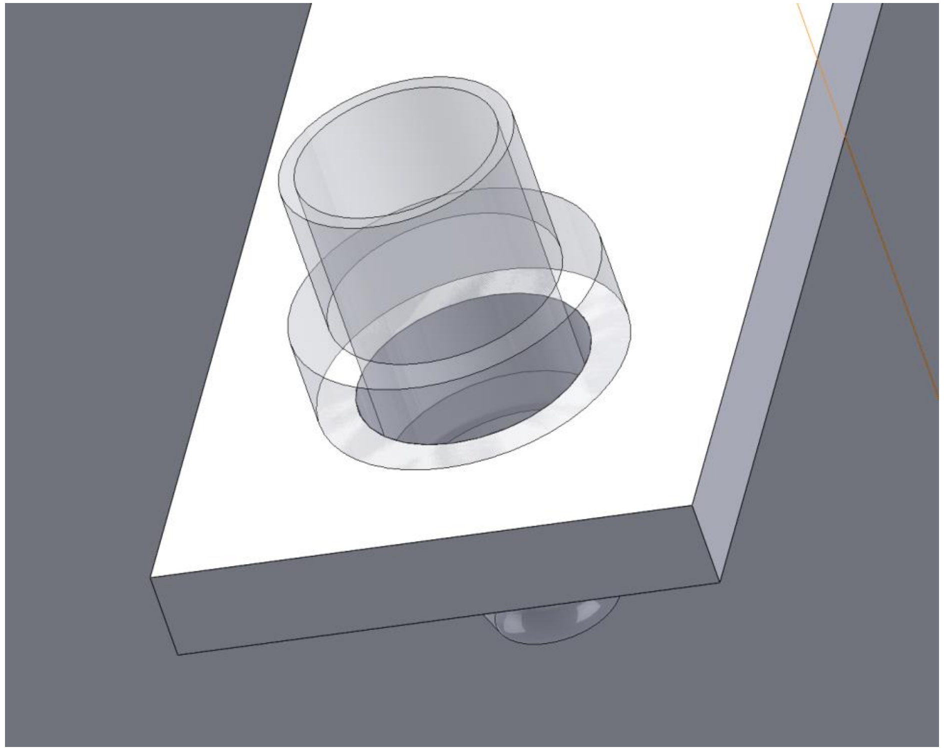


Maximum displacement here 2.8 mm at the end of the bucket (highlighted in red). Additionally, we observed that the adjacent buckets, situated close to the falling bucket, also experienced approximately 1.5 millimeters of displacement.

If we fail to stop it promptly and the water continues to fall for an extended period, there's a chance that the bucket may break.

Static Simulation of Nozzle Support

The nozzle has an external part that is connected to the support. The hole diameter is 15 mm, and the diameter of the external nozzle part is 20 mm. Let's calculate the area between them.



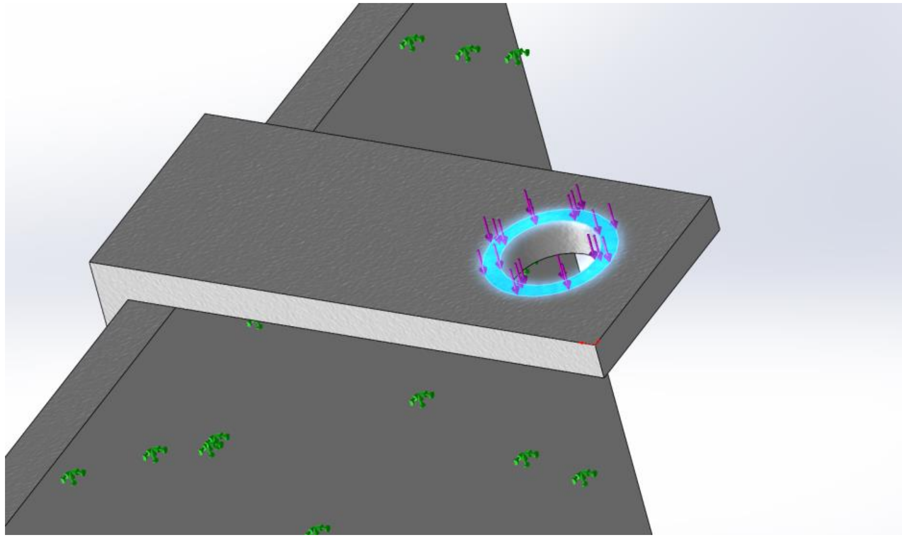
Lets, small circle is A1 and bigger circle is A2.
Formula of Area = πr^2, here r is radius of the circle.

Area of A1 = $\pi \times (7.5)^2$
Area of A2 = $\pi \times (10)^2$

Then, the area between the circles will be: A2 – A1
Area:
$A = \pi \times 100 - \pi \times 56.25$
$A = \pi \times (100 - 56.25)$
$A = \pi \times 43.75\ mm^2$
$A = 137.5\ mm^2$
$A = 0.0001375\ m^2$

So basicly the force will apply on this perticular area.

Now its time to think about how many Force is acting in this Particular area. We have Water velocity, Cross section area, Water density. Its our one of the challenge of our project. We have found some source that we can research about it.



One of our experience with,
$P = 0.5 \times \rho \times v^2$
here P is dynamic pressure.
Source:
https://en.wikipedia.org/wiki/Dynamic_pressure
Then we have formula, Force = Pressure x Area

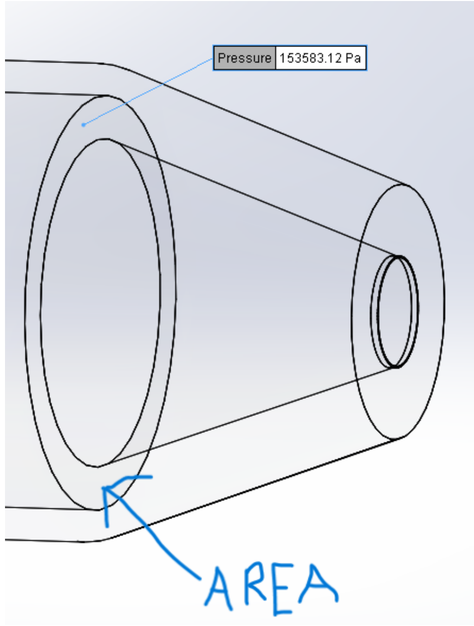
Here Water density is 1000 kg/m^3, Water velocity is 1 m/s, Cross-Section Area is 0.0001375 m^2.

$F = 0.5 \times 1000\ kg/m^3 \times 1^2\ m/s \times 0.0001375\ m^2$
We got result 0.07 N.

But the result with this formula that we got is not real life experience.

So, lets do flow simulation inside the nozzle and see how many pressure acting inside.

Within the nozzle, there is a level area where water will drop. In this case, water force will be applied. The external portion of the nozzle that is attached to the nozzle support will likewise experience this same level of force. Let's compute the force at work in this instance.



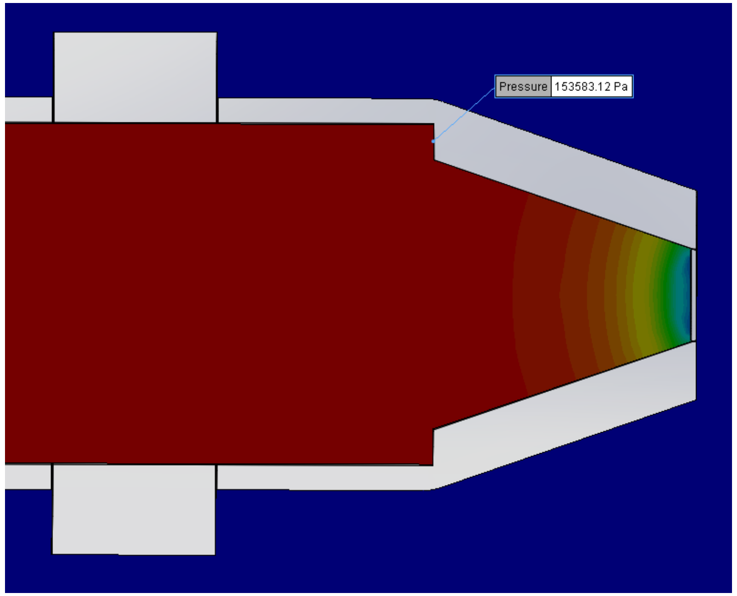
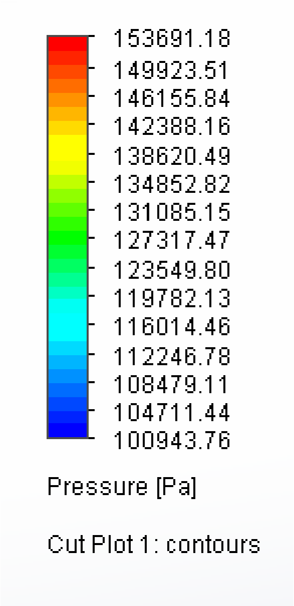


Fig 01



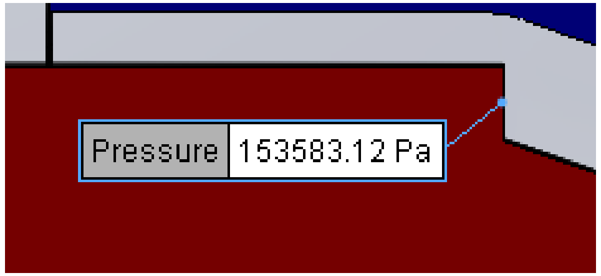
Diameter of the Bigger circle (B2): 13 mm
Diameter of the smaller circle (B1): 10 mm

Lets, calculate the area between them:

Area of B1 = $\pi \times (6.5)^2 = 132.7\ mm^2$
Area of B2 = $\pi \times (5)^2 = 78.5\ mm^2$

Then, the area between the circles will be: B2 – B1
Area is : $132.7 - 78.5 = 54.2\ mm^2 = 0.00005420\ m^2$

Simulation result: The water pressure is acting here **153583.12 Pa.**



Then we have formula, **Force = Pressure x Area**

Let's, calculate the Force,

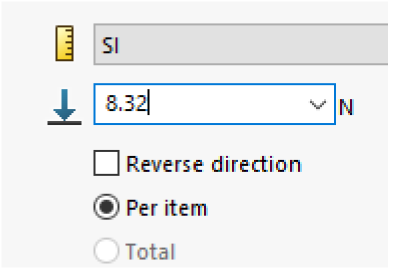
F = 153583.12 Pa x 0.00005420 m^2 = **8.32 N**

So, the force exerted by water flowing at a velocity of 0.7 m/s through a tap with a cross-sectional area of 54.2 mm^2 is approximately 8.32 N. This same amount of force will be applied on external support area 137.5 mm^2 because it is connected part. And It will apply on whole nozzle support. Now let's see how this force behave on the nozzle support.

Before the simulation, This data putted in the system to see the simulation:

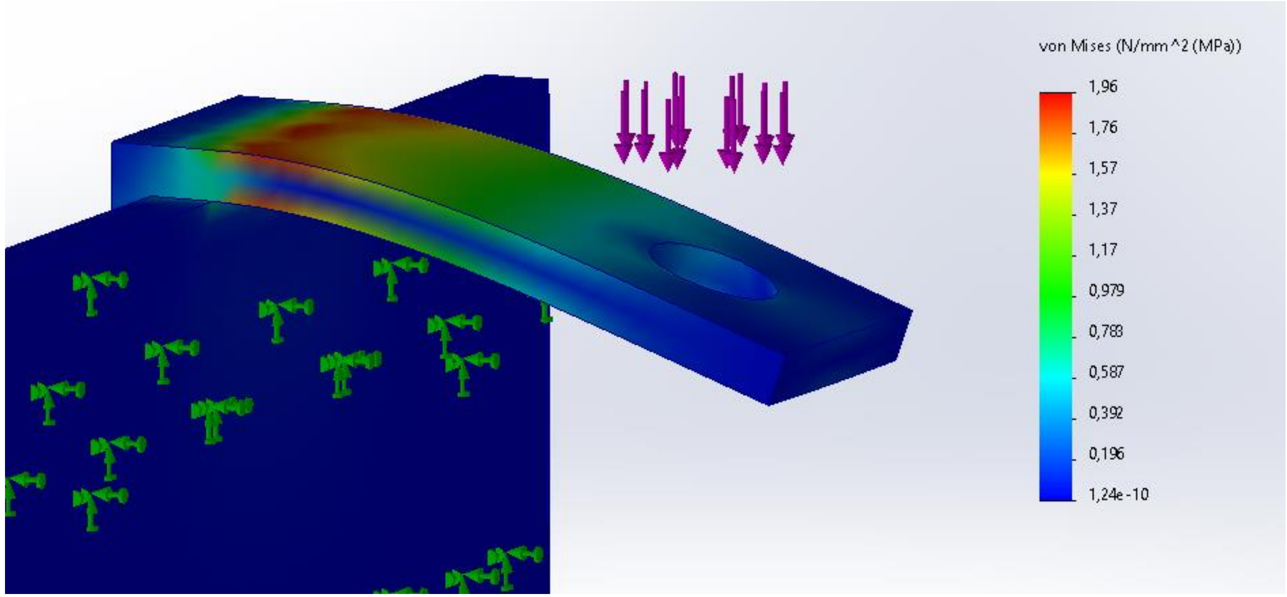
Material: ABS
Applied Force: 8.32 N

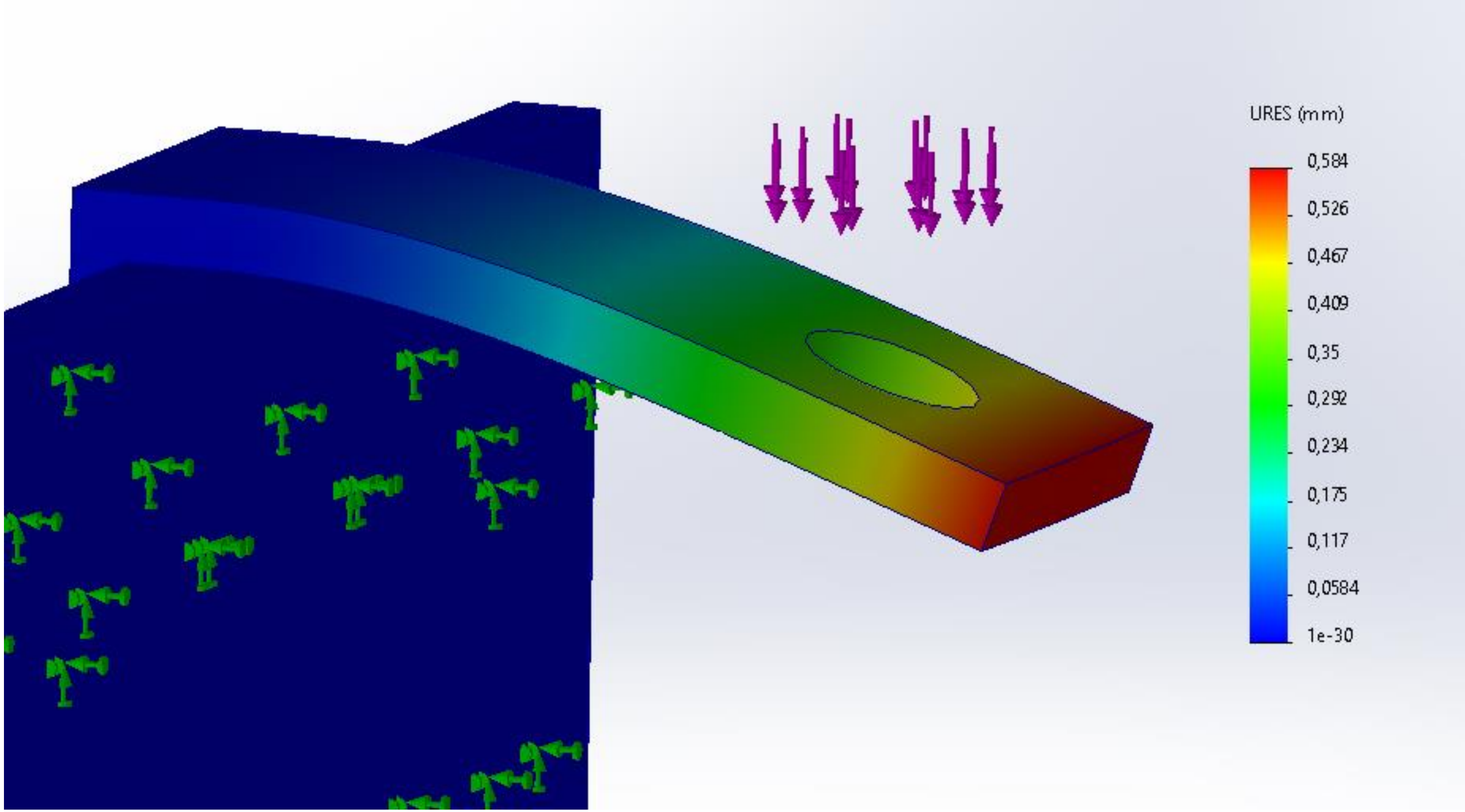
Note: Here used material PETG to build the prototype. There is no PETG option in SolidWorks simulation, so here used ABS which is very close to PETG.



Simulation Result:

The maximum stress observed is 1.96 N/mm², concentrated at the beginning (highlighted in red) of the component





Maximum displacement here 0.584 mm at the end of the part. 0.59 mm displacement is a quite small. So, The support can take the load of 8.32 N easily.

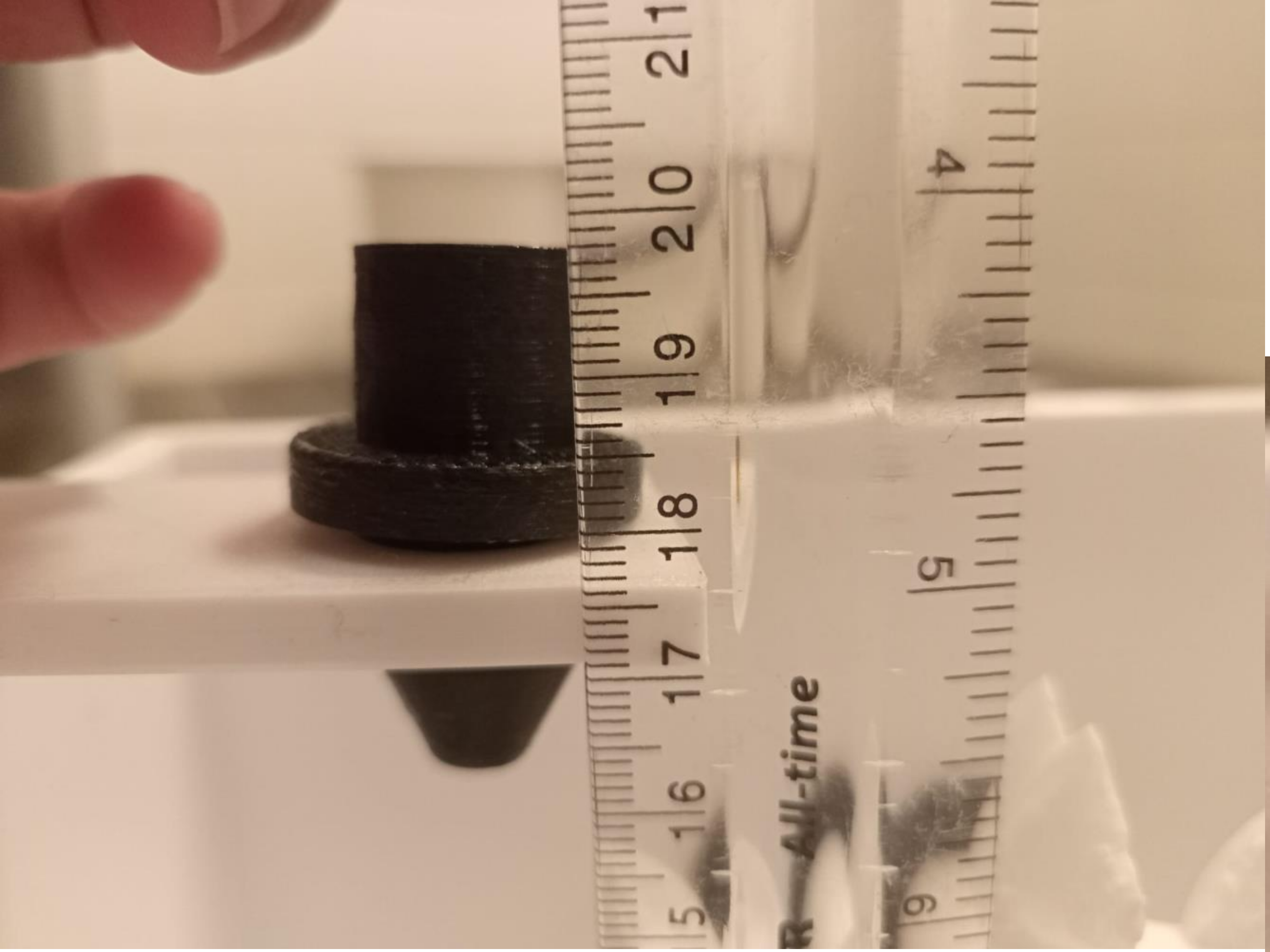


Fig 01 (left)

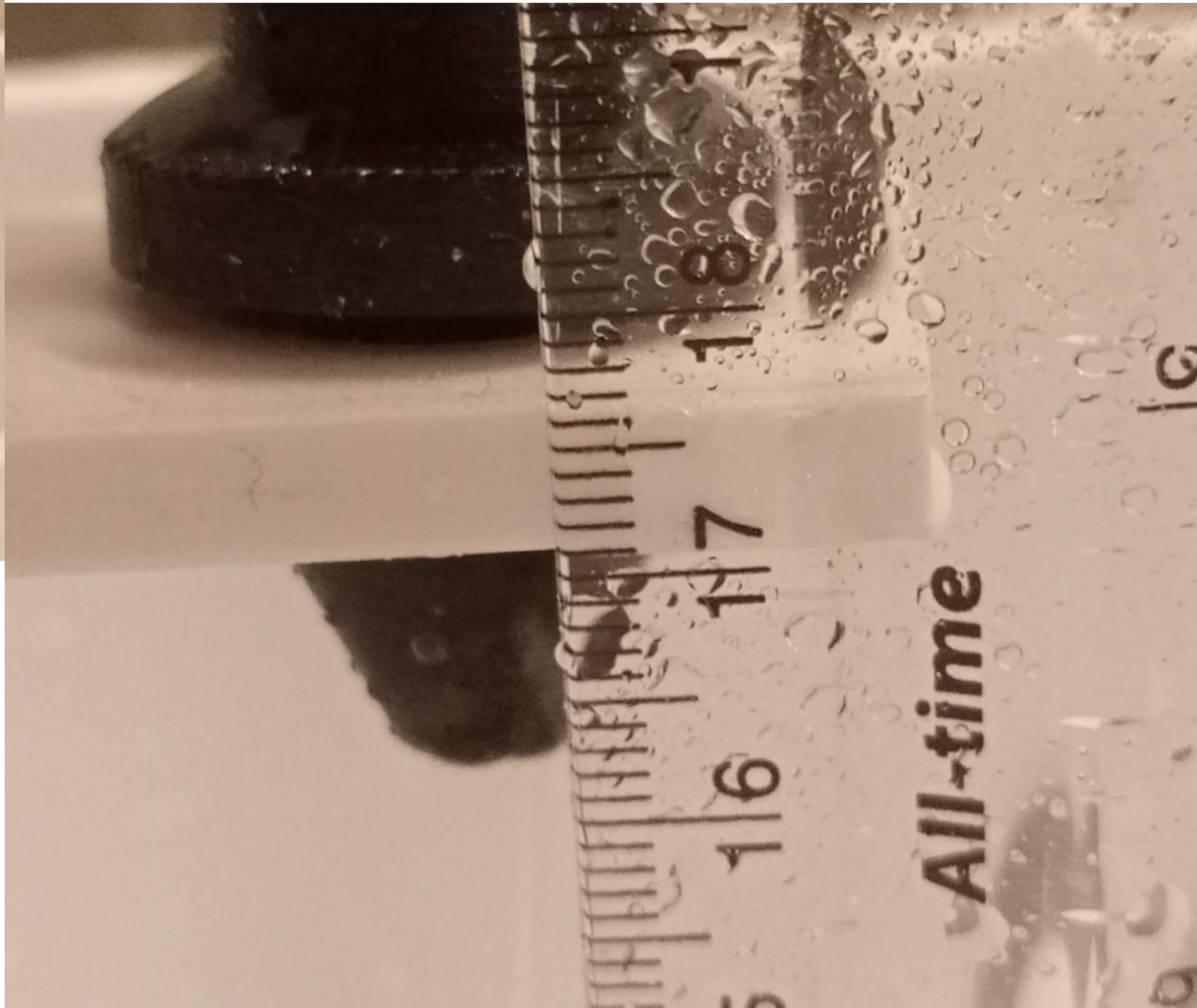


Fig 02 (right)

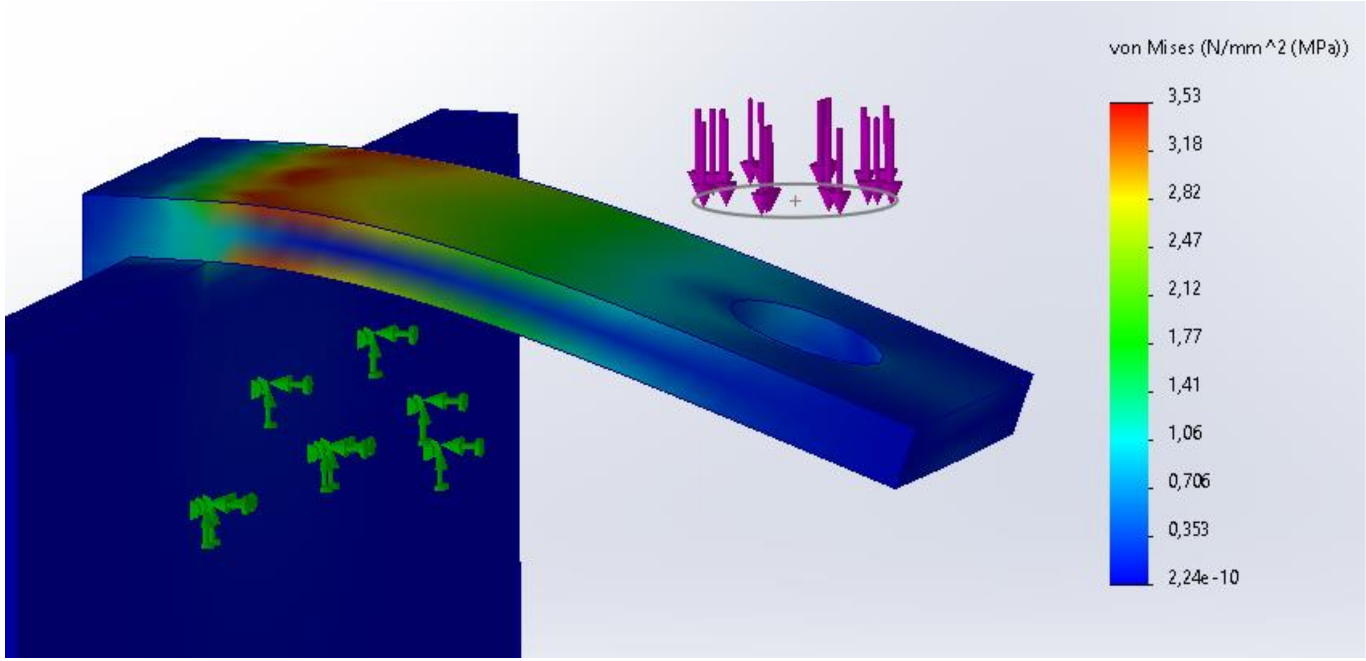
To ensure accuracy, we conducted real-world testing to assess the actual displacement. The simulation yielded a displacement of 0.584 mm.

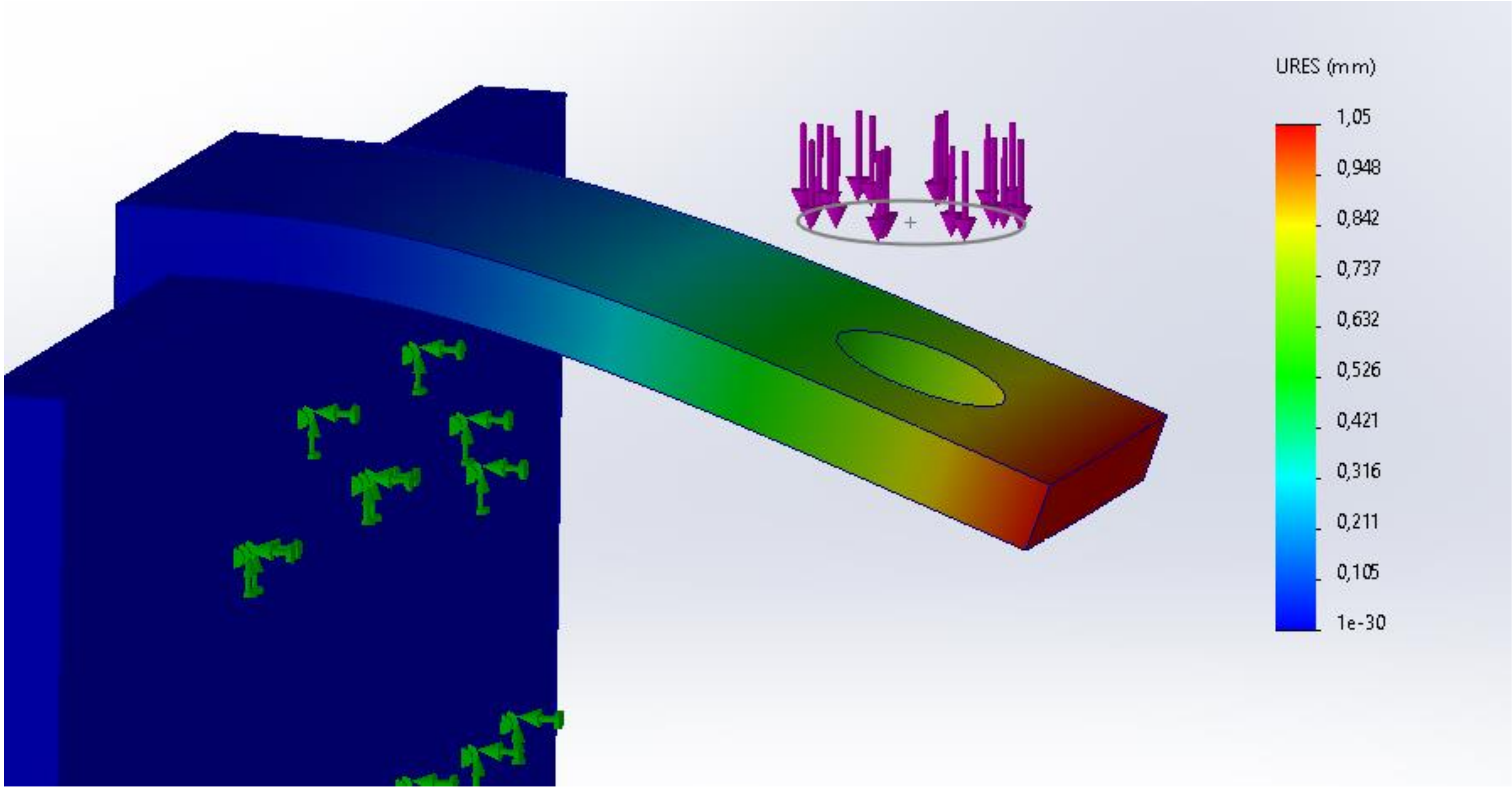
During the physical test, we initially measured the height of the nozzle support, confirming it as precisely 171 mm with corresponding markings on the miter scale (Fig 01). Upon observation under normal running conditions, the height remained consistent at 171 mm when viewed with the naked eye. However, upon closer inspection, a minute displacement was detected. Zooming in revealed a slight shift, with the nozzle support positioned slightly below the marked point. The measured displacement amounted to approximately 0.2 mm (Fig 02).

Lets test with 15 N load and see how it behaves

Simulation result:

The maximum stress observed is 3.53 N/mm², concentrated at the beginning (highlighted in red) of the component.





Maximum displacement here 1.05 mm at the end of the part. 1.05 mm displacement is a small but here water flow direction and angle from nozzle to wheel bucket is very important. The support can take the load of 15 N. **However, it is not good for more than this forward.**

For better result, Lets make a support (fig 01)
Now this time lets test with 100 N load. Lets see the simulation.

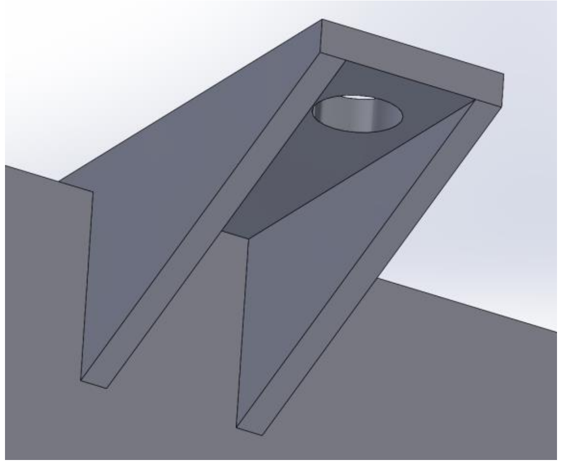
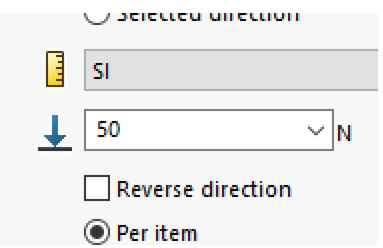


Fig 01



Simulation Result:

The maximum stress observed is 1.11 N/mm^2, concentrated at the beginning (highlighted in red) of the component. (fig 02)

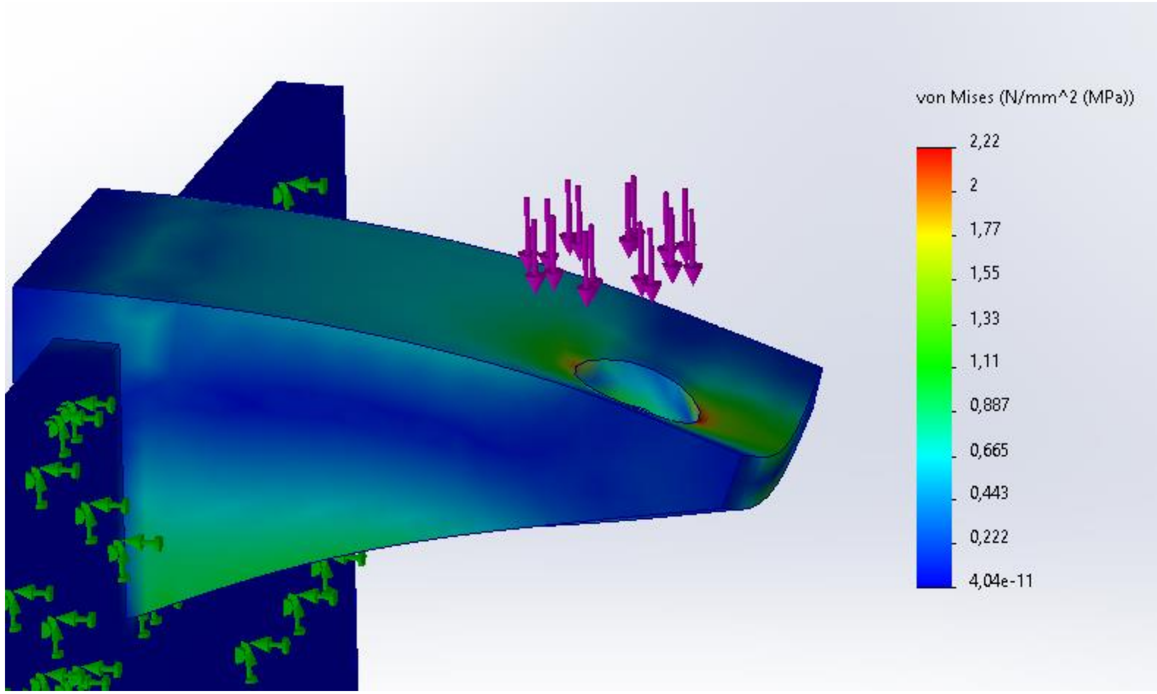
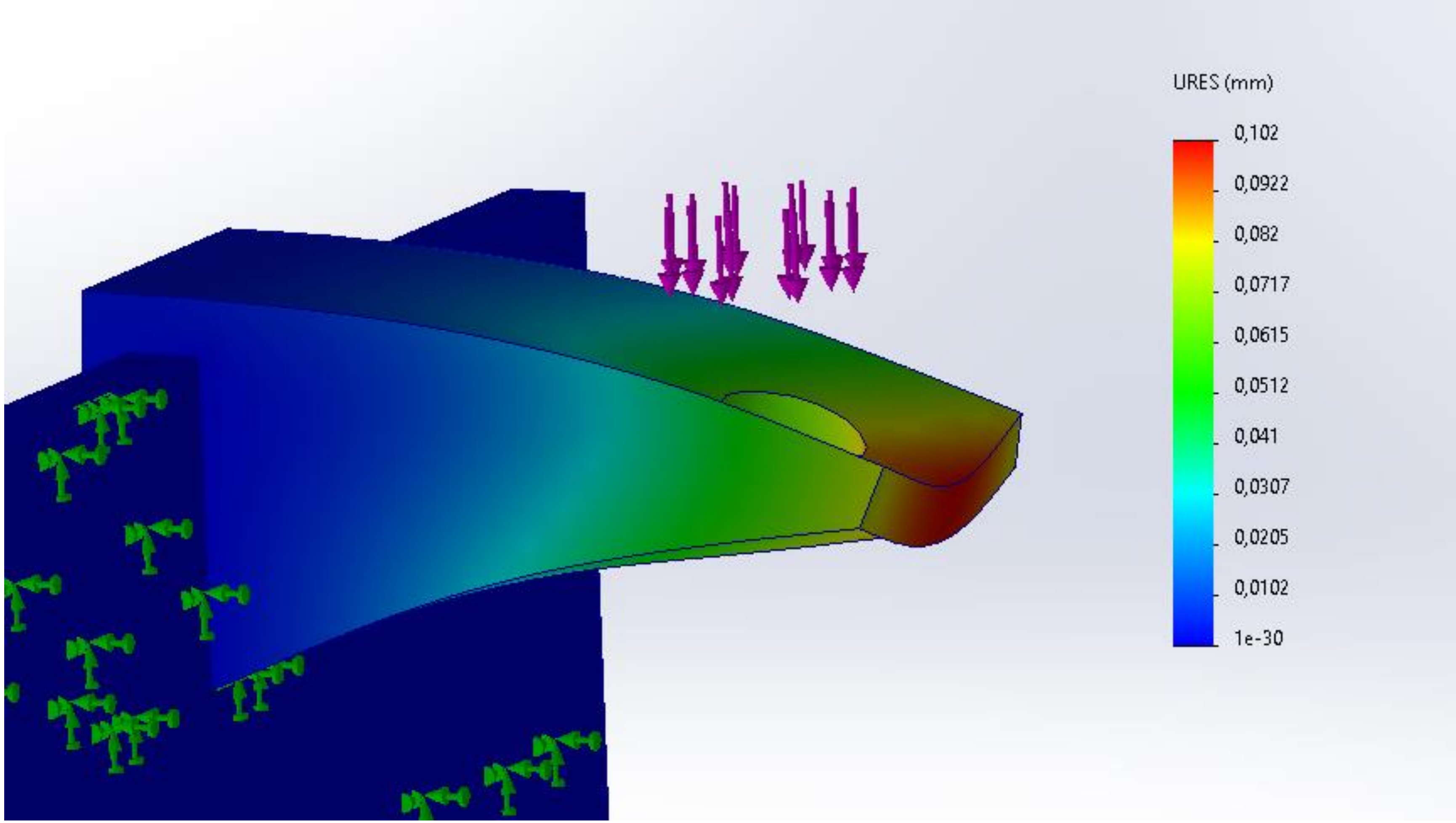


Fig 02



Maximum displacement here 0.102 mm at the end of the part. 0.102 mm displacement is a quite small. So, The support can take the load of 50 N Force easily.

3 Tools and lessons we have learned.

3,1 SolidWorks

3.1.1 Basic Designing

This project was an interesting way to learn more about the SolidWorks designing and a good way to use our skills in practice. Through different challenges, we have learned thanks to our professor and thanks to some external sources that almost everything is possible with SolidWorks. The first challenge was to draw the turbine (also called the wheel). It took us almost 3 days to achieve to finish it. We used many different functions like shell, circular repetition, 3D sketches etc…

Then another challenge and probably the most interesting one was to manufacture our own ball bearings. We just used little balls which were already manufactured and then we tried to print two ball bearings which could be used in our project. We decided to print it but before that we had to find a solution, the problem was the following one: How could we insert the balls into our bearing?

We decided to create a notch that allows the ball to be inserted when the two parts of the bearing are aligned. The idea was working but we still needed 6 tries to find decent settings.

A conclusion of it would be that even if you have good skills in designing, it is completely necessary to have a good understanding of the different systems you are handling.

3.1.2 Flow Simulation

One of our objectives is trying Flow Simulation on SolidWorks. SolidWorks Flow Simulation is a robust tool for analysing fluid behaviour in engineering projects. It handles various flow conditions and provides insights into velocity, pressure, and temperature distribution. Engineers can optimize designs, troubleshoot issues, and validate results with ease. With its wide application in industries like automotive and aerospace, SolidWorks Flow Simulation enhances design efficiency and performance.

The purpose of it is to simulate different setting related to the flow in function of our system. We decided to simulate the velocity, Pressure, and flow trajectory of our turbine with a flow which could be from the water tap.

3.2 3D Printing

We have learned different things about the 3D Topic. Before to learn how to print, it was interesting for us to learn the basic principle of it. The printer will make melt plastic (PLA, ABS, PETG etc…) and overlap many layers to give the desired shape. If we had to represent the process with steps:

- Design of your piece
- Save it as a STL file.
- Load your STL file in the slicer. The Slicer, in our case basically Prusa slicer could be considered as a link between your SolidWorks File and your 3D Printer.
- Apply the right settings, it's up to your printer, the plastic you want to use or the supports you might need. This step allows you to learn the duration of the printing.
- Then, export your file as a G-code which is suitable to be printed. Most of the time you'll have to export your g-code on a USB-Stick and then plug in in the printer.
- In our case there were no need to adjust any settings on the printer because everything was already done. But in some case, you would have to do the Support levelling or try different temperature in function of your plastic. There is only one parameter that we had to pay attention, it was to be sure that there was always enough filament. In our case we have used a Printer Prusa MK4 which is a basic printer.
- Then you just have to launch your printing
- If you needed support to print your piece, you will have to remote them after the printing.

3.3 Global comprehension of Mechanical concepts

This project enabled us to learn more and go deeper into plenty of subjects. Sometimes it is about things which could be considered as obvious, but it is always good and interesting to learn new things. We think that in our case the most important concept that we have better understood is probably the importance of tolerances. We always had to think about an appropriate tolerance and a way to assemble our different pieces. Most of the time we used embedding methods with very low tolerances, but we also used a key system. Then even if it did not work because of a lack of time, the understanding, and the study of the transmission system especially with gears has been an interesting and probably useful topic for our future. It would be a possibility to continue to work on this transmission system during the next semester. If we had to resume the things we have learned or better understood:

- Tolerances, a no skip concept which is the base of a lot of things related to engineering.
- Gear and belt system, how to transmit a movement.
- Importance of ball bearings in a system in rotation
- Fluid mechanics, use of Bernoulli and Continuity Equations for a real application

4 Peer Review

Steiner Lucas

Interesting Project with an intercultural group with student from different countries. Different methods of working but we overcame our communication difficulties to work in a more efficient way. It would have been interesting to have more time.

Mohammad Farhan Sadiq

Greetings, I'm Mohammad Farhan Sadiq, a participant in this group. I want to express my gratitude to all those involved in bringing this project to fruition. We are delighted because we have achieved making it happen. I'm delighted to share that my learning experience with this project has been enriching, allowing me to acquire valuable knowledge. Below, I'll briefly summarize the extent of my responsibilities and the value I've added to our collaborative efforts.

Generating Idea: In the initial stages, Lucas and I engaged in discussions regarding potential project ideas. Following meticulous research, we pinpointed this specific project concept and collectively agreed to move forward with its implementation. This iterative process not only deepened my comprehension but also allowed for the acquisition of invaluable insights pertaining to the project.

3D Printing: In this project, I took charge of the 3D printing aspect. my responsibility was printing every single component of the project. Each part underwent multiple printing iterations until achieving the desired functionality, such as four attempts only for the ball bearing. Then my responsibility was assembling them. This hands-on experience greatly enhanced my understanding of real-world tolerances, bridging the gap between theoretical knowledge from Jukka's class on 3D design and practical application. Prior to embarking on this project, I had limited knowledge of 3D printing. However, through dedicated research and hands-on practice, I gained proficiency in operating 3D printers, specifically the MK3s & MK4s, and mastering various 3D printing functions by project's end.

Simulation: In this project, I took charge of the flow Simulation and static simulation aspect. Utilizing both SolidWorks and Ansys, I conducted comprehensive analyses, obtaining noteworthy results that were included in the report.

Product Development: Initially, our first design for the nozzle positioned horizontally. However, we later explored the concept of utilizing gravity by switching the orientation of the nozzle to vertical. I bought a water pipe from MiniMani shop and modified our nozzle design to ensure it matches our water pipe perfectly. Though the initial design for our nozzle support failed, we refined it until it proved effective. Upon conducting our first round of tests, we observed poor water distribution. As a result, we iterated on the box design by introducing a cover and incorporating a one-way exit to address this issue.

Summary: Through active participation in this project, I have not only expanded my practical skills in 3D printing, simulation, and product development but also developed a deeper understanding of collaborative problem-solving and iterative design processes. This experience has been invaluable in bridging the gap between theoretical knowledge and real-world application. We acquired extensive knowledge through our collaboration with the team. We've gleaned invaluable lessons from this experience, moving forward we are eager to continue applying these newfound skills and insights to future projects, further contributing to our collective success.

Rajiv Rahut

Hello, my name is Rajiv Rahut one of the members of this group as I would like to thank our team leader Lucas for giving us so much support and guidance that we need to complete our project. All of our group members have their parts to do and I think we all had contributed some things that's was needed on this project.

In this project I learned to design the bearings thats feet's on the shaft, it was Pretty hard for me to design it because I have to design the bearing according to the shaft size and the ball that is fitted inside the bearings because it was made of plastic and we had that ball from JHC so it was pretty exciting although we printed 2-3 times to get exactly the same size.

I also learned to calculate the whirling of shaft like critical speed we are in modern generation so because of the internet we can get easily the ideas that we need and I did it but anyway I have some good ideas to calculate and to design the bearings that I had learned from this project. We also learned that how to communicate and create different ideas to make sure that we have the same goals.

MD Al Amin

This is Md. Al-Amin. During the Pelton turbine project, I used a thorough approach, focusing on calculations, testing, and design. Successes were achieved by optimizing parts like the nozzle and lightbox, utilizing Bernoulli's equation for efficiency. Testing offered insights into continuous improvements. Initially, difficulties in reconciling theoretical calculations with actual results developed, requiring a more in-depth investigation of fluid dynamics concepts. This helped me develop a better grasp of turbine design difficulties and the need for extensive validation processes. Overall, the project provided us with great lessons about interdisciplinary teamwork, issue resolution, and iterative design optimization.

5 Conclusion

While this project encountered some limitations compared to our initial expectations, we're proud to declare it a success. It has fulfilled its primary objective of generating electricity as intended. We have learned plenty of things and improve our skills as future engineers. The best way to learn is to do mistakes and we have done plenty. From our ball bearings which were too loose to our nozzle which was not fitting with the tap, we have been through a lot of different problems, and we solved most of them. We'll know how to remedy this problem in the future. We are convinced that this lecture enabled us to learn way better than with a normal lecture. We have learned things because we really needed it and we are sure that when you are learning that way, you won’t forget it. If we had to do this project again, it would be for sure better, and this is already a prove that we increased our skills. Then, even though the engineering aspect of our project was the most relevant point, we too had to learn to work as a group, listen to each other and put ourselves in question. In conclusion, this project gave us as much in terms of engineering as it did in terms of group work. A lot of things didn't work out as planned, but the feedback is still very positive. Apart from this project, the fact that we were often at the workshop enabled us to meet a lot of competent people who taught us an enormous number of skills. We didn't use it for this project, but thanks to them we learned how to weld and use a CNC.

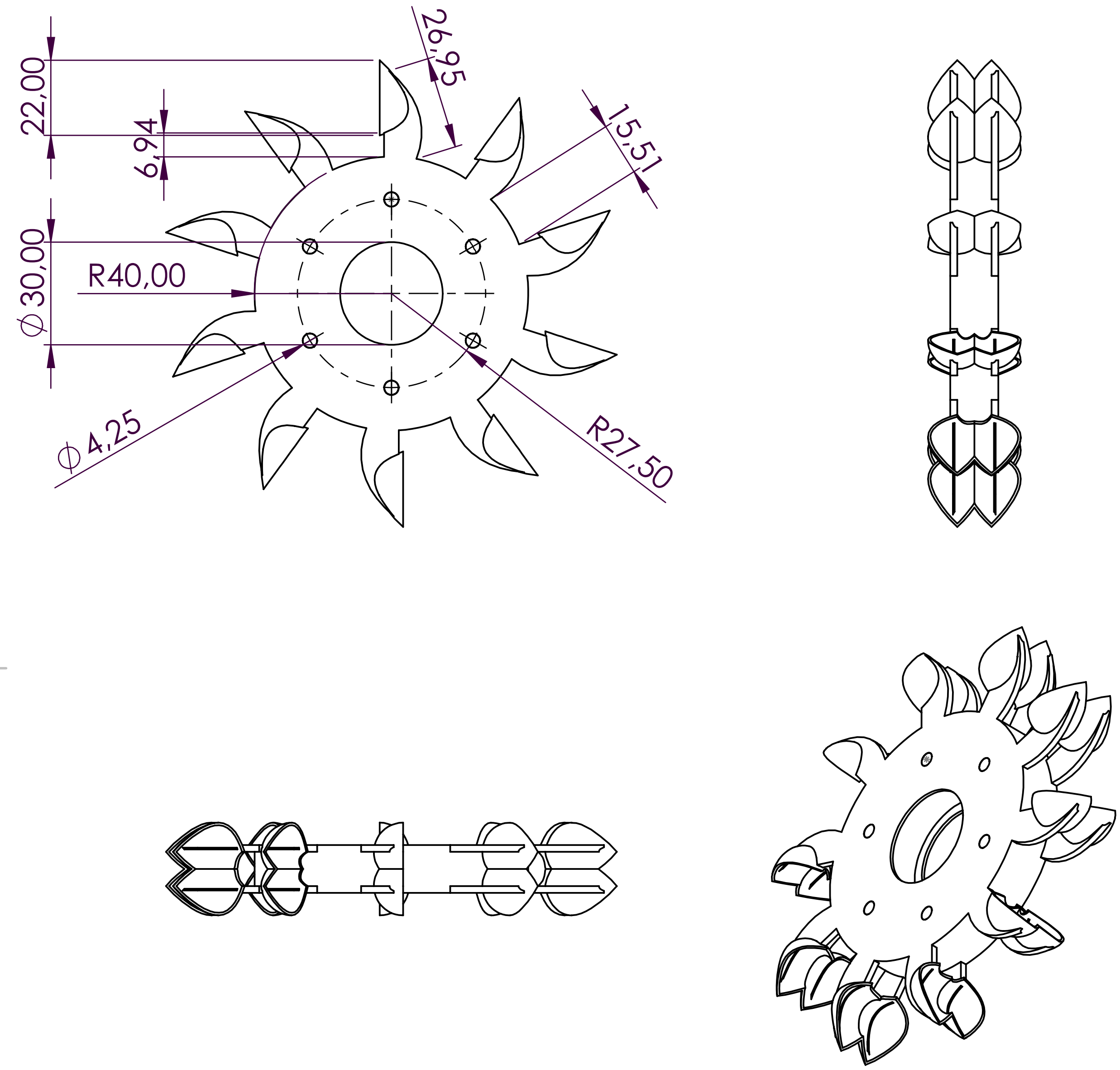
Sources

Figure	Source
Figure 1: Representation of a Pelton Turbine	https://www.researchgate.net/figure/Principal-scheme-of-the-Pelton-turbine_fig1_336931385

References

Tyler, B. 2019. Example source name. Diamond Education e-books. Dublin: InfoPoint Publishing.

Virtanen, V. 2021. Example source title. LAB University of Applied Sciences. Retrieved on 1 January 2022. Available at http://example.fi



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE:

MATERIAL:

DWG NO.

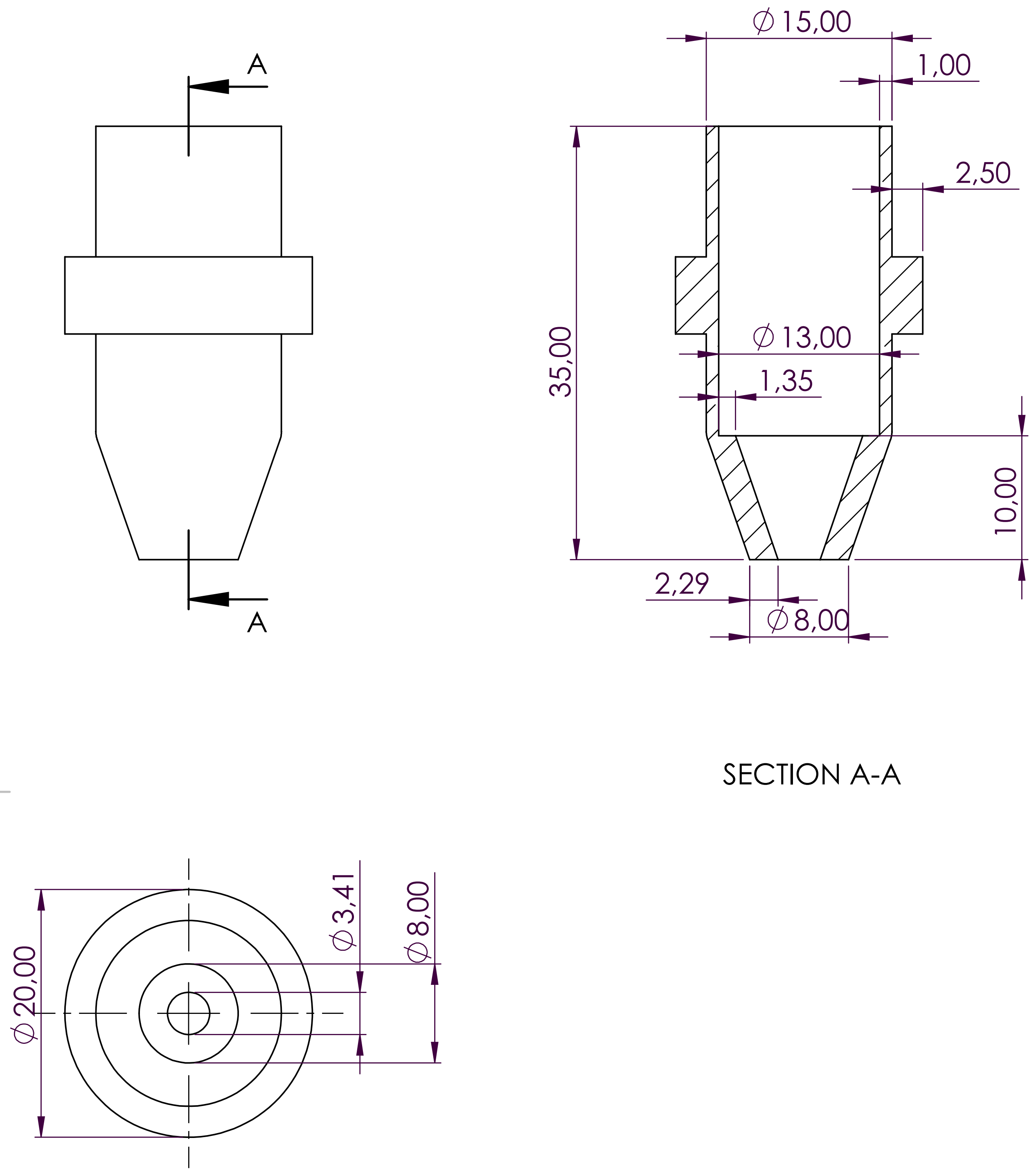
wheel

A4

WEIGHT:

SCALE:1:2

SHEET 1 OF 1



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DEBURR AND BREAK SHARP EDGES

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MATERIAL:
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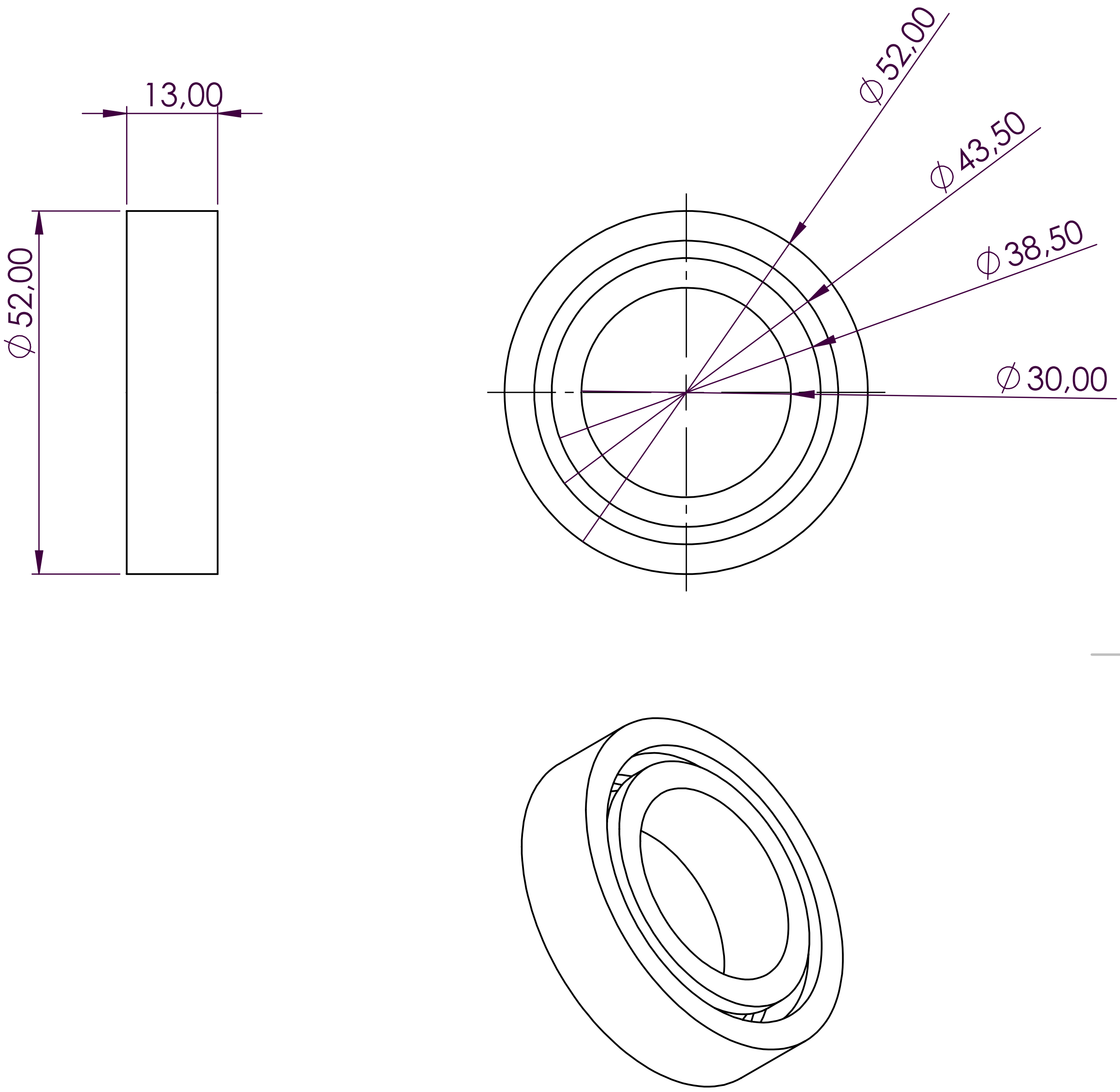
SHEET 1 OF 1



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DEBURR AND
BREAK SHARP
EDGES

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MATERIAL:
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WEIGHT:

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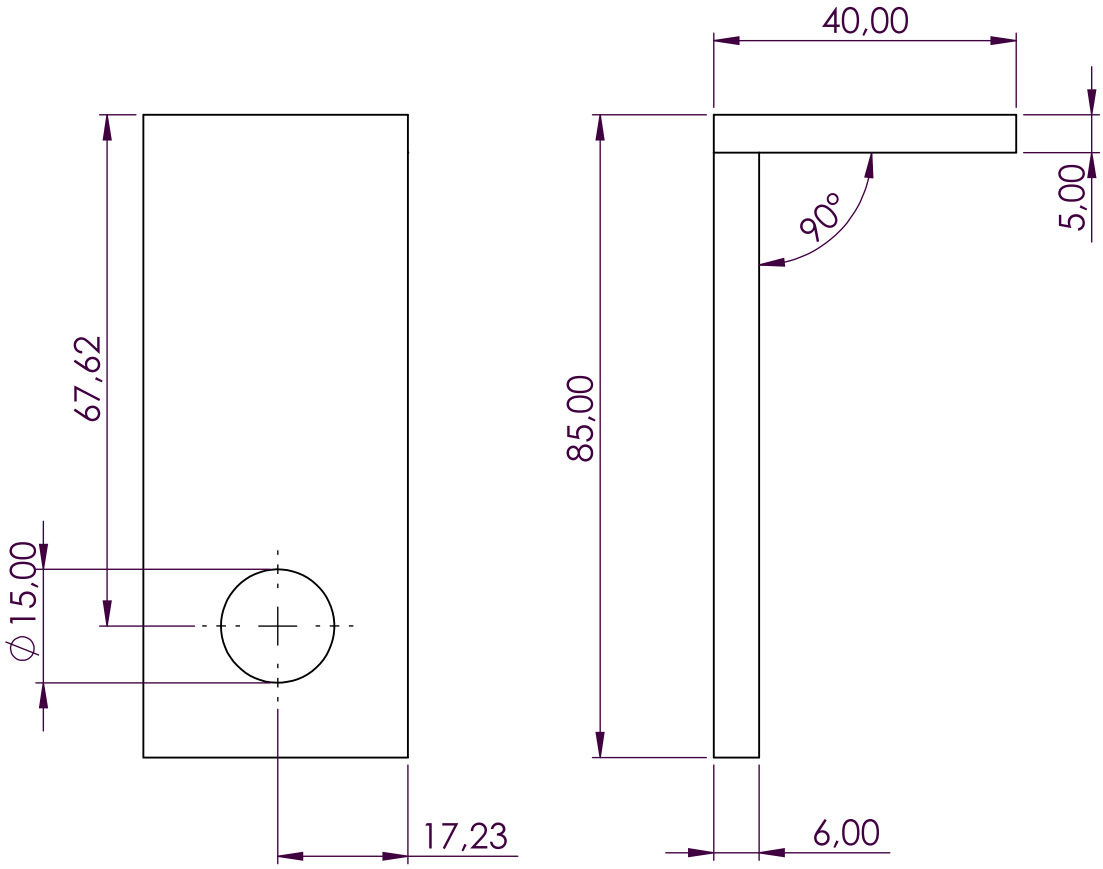
SHEET 1 OF 1



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FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE:

MATERIAL:

DWG NO.

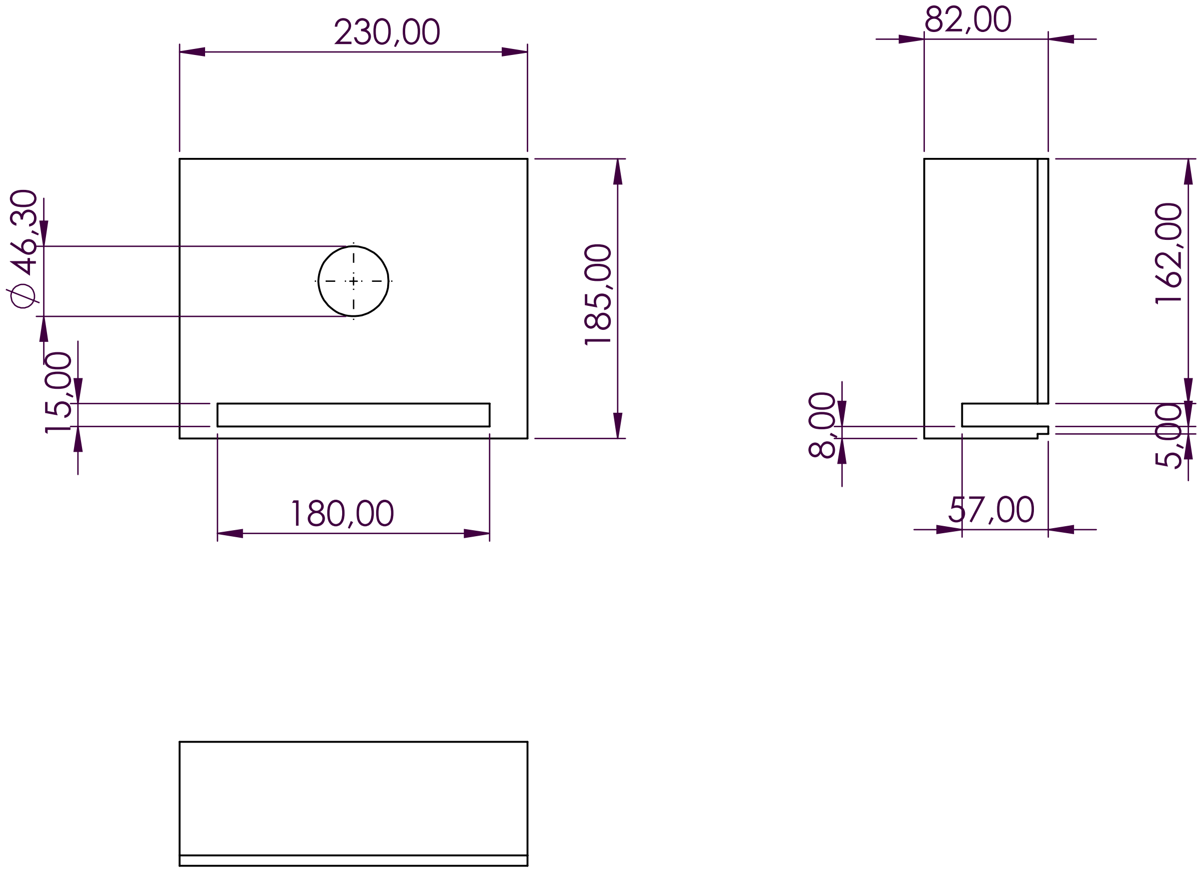
nozzele support

A4

WEIGHT:

SCALE:1:1

SHEET 1 OF 1



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:	FINISH:	DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING / REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE:

MATERIAL: Material <not specified>

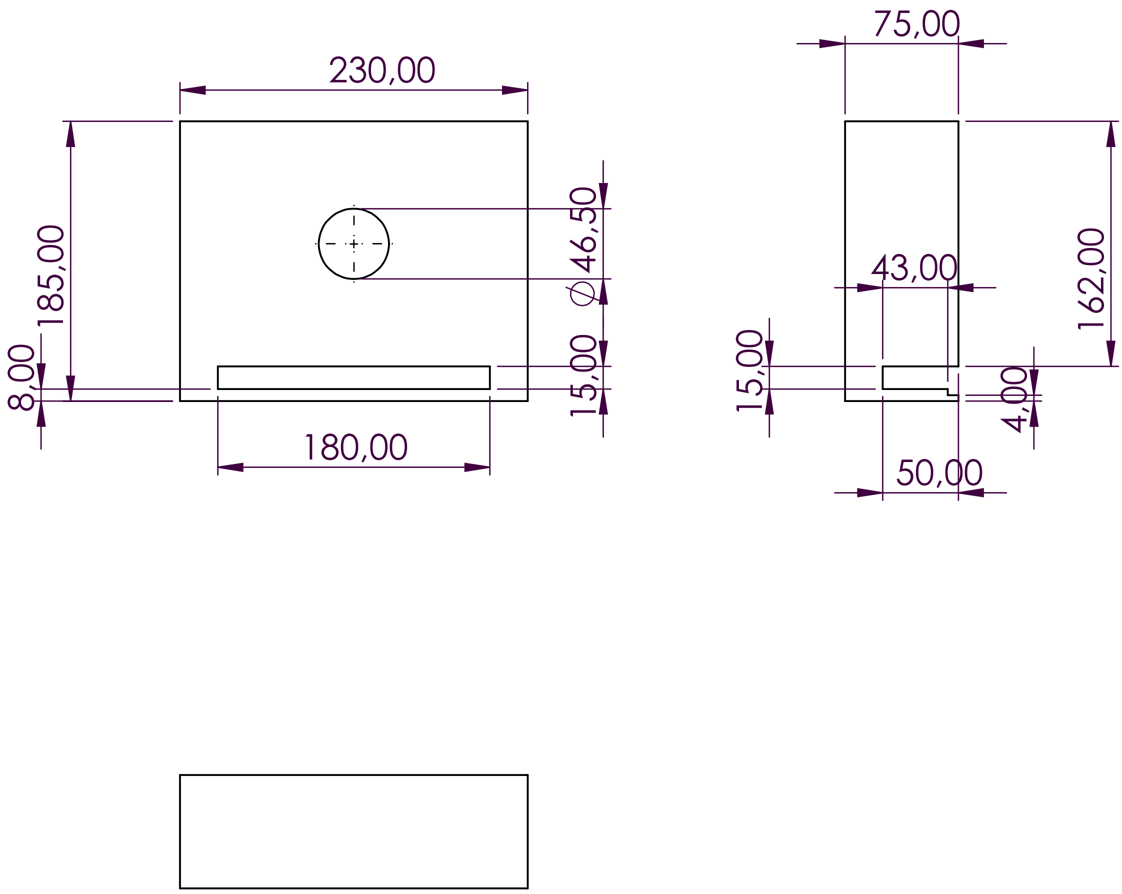
DWG NO. half box 1

A4

WEIGHT:

SCALE:1:5

SHEET 1 OF 1



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE:

MATERIAL:
Material <not specified>

DWG NO.
half box 2

A4

WEIGHT:

SCALE:1:5

SHEET 1 OF 2